



**Verified Carbon
Standard**

DELAGUA IMPROVED COOKSTOVE GROUPED PROJECT



DOCUMENT PREPARED BY DELAGUA

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Summary description of the technologies/measures to be implemented by the project:

The clean cooking grouped project 3699 is a voluntary Grouped Project (GP) aimed at distributing fuel efficient improved cookstoves (ICS) to households in Rwanda. The Project Participant (PP) of the GP for all the communication, management, and ownership rights of the Greenhouse Gas (GHG) Emission Reduction Units (ERU) is DelAgua Trans Africa Stoves. Pte. Ltd. (DATAS) and the ICS under this GP will be distributed by DATAS. The distribution of ICS is not mandated by Rwandan law and the Grouped Project is a voluntary initiative being run by the PP. The program is not being implemented as a result of any government legislation or intervention. Throughout this Project Description, this grouped project may be interchangeably referred as “program”. The PP plan to distribute 500,000¹ improved cookstove under the grouped project during first renewable crediting period. It is intended that under this GP two types of single pot portable stoves will be distributed, which are SSM S32 X stove and Burn Kuniokoa G2. Each household will receive 1 ICS.

The location of the project:

Under the GP, PP will distribute the project technologies to individual households and communities within the host country Rwanda. The PAIs will occur in geographically defined regions within Rwanda. The details of the GP and PAI locations are provided in Section 1.12 of this document.

An explanation of how the project is expected to generate GHG emission reductions or removals:

The VCS GP will reduce the firewood usage at household level, thereby reducing the GHG emissions compared to the baseline. The fuelwood savings will act as the main source of VCU generation for this project. This fuelwood savings is estimated for the crediting period and calculated for the first monitoring period using the VCS approved methodology VMR0006, ver 1.1. The GP will follow the VCS Methodology from Sectoral Scope 3 - VMR0006 “Methodology for Installation of High Efficiency Firewood Cookstoves”, version 1.1 for ER calculations. The implementation (distribution of ICS units) and parameters set for ER calculations within the GP will be monitored on a periodic basis, based on the requirements provided within the applied methodology.

¹ The initial distribution projections for the first crediting period were 250,000 however PP was able to distribute 130,633 ICS free of charge in less than two months. Therefore, the investment was increased to facilitate up to 500,000 ICS to ensure the project could reach and benefit as many households as possible.

A brief description of the scenario existing prior to the implementation of the project:

The scenario existing prior to the implementation of the project activity would be usage of 3 stone fire and traditional stoves, which have a very low thermal efficiency, and therefore consume a higher amount of firewood as fuel for cooking. The applied methodology recognizes this efficiency value for traditional baseline devices without any flue gas ventilation (such as grate or chimney) at 0.1 (or 10%). On the other hand, the project device being distributed to households have a high thermal efficiency of 51%² with pot-skirt.

An estimate of annual average and total GHG emission reductions and removals:

The average annual and total GHG emission reduction expected from the grouped project is expected to be 2,149,016 and 17,192,130 tCO₂e, respectively, over the first 7-year crediting period.

<u>Audit Type</u>	<u>Period</u>	<u>Program</u>	<u>VVB Name</u>	<u>Number of years</u>
Joint Validation & Verification	04-November-2022 - 31-December-2022	Verra VCS	Earthood Services Private Limited	0.158 year (58 days)
<u>Total</u>	<u>1</u>	<u>1</u>	<u>1c</u>	0.158 year

1.2 Sectoral Scope and Project Type

Sectoral Scope 03 – Energy demand

Type II – Energy efficiency improvement projects

Activity category – non AFOLU

The project activity is designed as a Grouped Project Activity (GP) and includes multiple Project Activity Instances (PAIs) of the same activity type into one Grouped Activity. New instances can be introduced to the GP at any monitoring period, with a PAI defined as “Distribution of efficient cookstoves to a local community within a specific geographic and climatic region in the project boundary”.

1.3 Project Eligibility

² Please refer manufacturer’s specifications for Burn Kuniokoa gen. 2 model to verify this value.

The Grouped Project Activity falls within the scope of the VCS Program. This is elaborated under the tables below.

Table 1: VCS Program scope

Scope	Applicability
The seven Kyoto Protocol greenhouse gases.	The GP reduces GHG emissions associated with the combustion of cooking fuels. Gases included are CO ₂ , CH ₄ , N ₂ O.
Ozone-depleting substances (ODS).	The GP does not involve any ODS.
Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	The ICS to be distributed in the GP applies the methodology “VMR0006 “Methodology for Installation of High Efficiency Firewood Cookstoves”, version 1.1.” which is a VCS approved methodology.
Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval.	Not applicable.

Table 2: Excluded project activities under the VCS Program

Excluded Activity	Applicability
Grid-connected electricity generation activities using hydroelectric power plants.	N/A The GP reduces GHG emissions associated with the combustion of cooking fuels.
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants.	N/A. The GP activities are energy efficiency measures and do not generate electricity from wind, geothermal, or solar power plants/units.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration.	N/A The GP activities are energy efficiency measures and do not utilize recovered waste heat for any purposes.

Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass.	N/A. The GP activities are energy efficiency measures and do not generate electricity using biomass. The GP includes energy efficiency improvements in thermal applications, i.e. cookstoves.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel.	N/A. The GP activities do not include the generation of electricity and/or thermal energy using fossil fuels, including activities that involve switching from a higher carbon content fuel to a lower carbon content fuel.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs).	N/A The GP activities do not include the replacement of electric lighting.
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers.	N/A. The GP activities do not include the installation and/or replacement of electricity transmission lines and/or energy efficient transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	N/A. The GP reduces GHG emissions associated with the combustion of cooking fuels

1.4 Project Design

The project has been designed to include distribution of ICS units at multiple locations (districts, sectors, cells, and umudugudus) within the project boundary, under multiple project activity instances within a grouped project.

Eligibility Criteria

VCS 3699, being a Grouped Project Activity, will be implemented while fulfilling the eligibility criteria provided in the Table below.

Eligibility criteria as per section 3.6.10 to 3.6.15 of the VCS Standard v4.4

Criterion	How the new project activity instances comply	Evidence
Grouped projects shall specify one or more clearly defined geographic areas within which project activity instances may be developed. Such geographic areas shall be specified using geodetic polygons	Each PAI will have clearly defined geographic areas and such area will be specified using geodetic polygons.	PAI 1 has been implemented in Rwanda. PP has submitted geodetic polygons to VVB and added details in section 1.12 of the PD MR.
Determination of baseline scenario and demonstration of additionality are based upon the initial project activity instances. The initial project activity instances are those that are included in the project description at validation and shall include all project activity instances currently implemented on the issue date of the project description.	Baseline scenario and demonstration of additionality shall be based upon the initial project activity instances	Baseline scenario and additionality has been demonstrated for PAI1 which is included at the time of validation in section 3.4 and 3.5 of the PD MR.
Where a grouped project includes multiple project activities, the project description shall designate which project activities may occur in each geographic area.	The grouped project includes only one project activity.	The PAI-1 includes distribution of improved cookstoves to households that are dependent on traditional cookstoves using woody biomass for cooking.
The baseline scenario for a project activity shall be determined for each designated geographic area, in accordance with the methodology applied to the project.	Baseline scenario for a project activity shall be determined for each designated geographic area, in accordance with the methodology applied to the project	Baseline scenario has been determined for PAI-1 for Rwanda. Details have been added in section 3.4
The additionality of the initial project activity instances shall	Additionality for a project activity shall be determined	Additionality has been determined as per the

be demonstrated for each designated geographic area, in accordance with the methodology applied to the project	for each designated geographic area, in accordance with the methodology applied to the project	designated methodology. Details have been added in section 3.5.
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Eligibility criteria as per section 3.6.16 and 3.6.17 of the VCS Standard v4.4:

S No	Criterion	How the new project activity instances comply	Evidence
1	Project Activity Instances (PAIs) must meet the applicability conditions set out in the applied methodology.	Please refer the subsections below.	Please refer the subsections below.
1a	Project activities shall be implemented in domestic premises or in community-based kitchens.	Each PAI will distribute technology that is appropriate for domestic use to household end-users	a) Manufacturer's specifications. b) Project ICS distribution database containing unique household end user information. All the ICS under PAI-1 distributed to individual households, the distribution records with end users details shared with VVB for their reference.
1b	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove	ICS distributed in each PAI will have a thermal efficiency of at least 25%, and will be confirmed by the end user that they were using the firewood as the cooking fuel in the project's baseline.	1.Manufacturer's specifications. 2. Information recorded within the baseline KPT and project monitoring surveys on the fuel usage. For PAI-1, lab test efficiency certificate for model distributed under this instance has clearly

	replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	Each PAI will record the baseline and project fuel use to ensure that wood fuel is the sole source of energy.	showing the efficiency above 25% is being submitted to VVB. The distribution record and sample survey results submitted to VVB, which includes the baseline fuel usage i.e. fuel wood by all end users under PAI-1.
1c	Both 'Projects' and 'Large Projects' can use the methodology	PAIs will comply with the GP requirements	The GP is defined as a "Large Project" according to the VCS Standard ver. 4.4 requirement in Section 3.10.1, where the annual estimates of the VCUs is greater than 300,000 tonnes of CO _{2e} per year. No further assessment required at PAI level.
1d	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	The PP has demonstrated at grouped project level that Non-renewable biomass has been used in the project region since 31 December 1989 under section 3.2.	No further assessment required during inclusion of PAI.
1e	For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one	Biomass residues are not processed as a fuel to be used in the GP. End users will be asked to state their source of energy and baseline appliance at the point of distribution and during the project monitoring.	1. Monitoring surveys. 2. Project ICS distribution poster specific to the end-users (sample photographs) The distribution database and sample survey record are being shared with VVB, which clearly shows that fuel used by end

	type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	Above information will be stored in the monitoring database.	users under project activity are not processed.
1f	The Project Design Document shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo)	To avoid double counting of emission reductions through the method of distribution, each cookstove has a unique identifier number and the distribution of each stove is logged onto a database which includes unique information for each recipient Each PAI will have system for avoiding double counting with data being collected via a smartphone app (where possible), or via paper forms, including: 1. Date of distribution	1. Project (GP) Distribution database can be checked to verify the distribution approach, along with the distribution system (includes both plan and execution steps) established by the PP. The distribution database is an electronic document system. 2. The VCS GP ICS units will have unique engravings or text specifically mentioning the unit is strictly “not for sale” along with the DelAgua logo. For PAI-1, the distribution record and written attestation from the

		2. A unique serial number³ on each technology 3. Available end-user data: name, phone number, address/physical location, ID number	project proponent confirming that the ICSs distributed are not part of any other CDM or Voluntary GHG programme is provided to VVB.
1g	The Project Design Document shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The End User Agreement will confirm that the ownership rights of the ER/Vcus generated by the PAI reside with the PP. The signature of each end user confirming their understanding of this agreement will be collected at the point of distribution. The End User Agreement will be in English and translated into a local language where required. If necessary, it will be read out to the End User. All rights to distribute the delivered stoves and claim the resultant GHG emissions reductions belong to the PP. The GP will not include any PAI that has been rejected by another GP, or PA-DDs of any other GHG programs.	a) End users will be asked to sign the End User Agreement and this information will be a part of the agreement, and then stored in the monitoring database. b) Each PAI will include a statement on the uniqueness of the Project and by listing other similar activities (from other GHG programme registries) that are taking place within the project boundary to demonstrate its uniqueness.

³ There are no similar programs running in the host country Rwanda where the same ICS models are distributed by another entity under different program(s). If this is found during the project ICS monitoring or verification, it shall be ensured that double counting is avoided by using distinct nomenclature for the unique ID code of the ICS compared with the different programs, and also making sure that it has a distinct color, label, or marking.

2	Use the technologies or measures specified in the project description.	Only ICS that conform with the GP description are to be distributed in the project. The ICS will be chosen to deliver a level of service at least equivalent to the baseline appliance.	Manufacturer's specifications. The ICS deployed under PAI-1 is Kuniokoa Generation 2 with Pot Skirt, which is same as described in section 1.11 of JPD/MR.
3	Apply the technologies or measures in the same manner as specified in the project description.	The ICS distributed in the PAIs will adhere to the GP description (i.e. burning of biomass for cooking meals) and replace the baseline device which also was utilising the biomass burned in the baseline scenario.	Manufacturer's specifications. The ICS distributed under PAI-1 reduces fuel wood consumption due to better thermal efficiency and replaces the baseline traditional stoves, the same can be checked with distribution record and sample survey results.
4	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	All new PAIs will be implemented only in regions within the geographic borders of Rwanda subject to the same baseline scenario determined in Section 3.4 in this Project Description. The geographical location of the project ICS can be traced through the information provided in the distribution database, which includes the location data of the ICS distributed.	ICS Distribution database For PAI-1, records distribution (end user registration database) with clear delineation of project boundary i.e. host country of Rwanda is provided to the VVB for their reference.
5	Have characteristics with respect to additionality that are consistent with	All new PAIs will use the activity method for	Step1: Regulatory Surplus

	the initial instances for the specified project activity and geographic area.	demonstration of additionality. Step 1: Regulatory Surplus There is no government mandated programme or policy in host country of this project ensuring the distribution of new project activity instances. Step 2: Positive List The inclusion of new project activity instances will comply with the positive list as it satisfies criterion 1 where it meets all the applicability conditions of the methodology. Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.	There is no mandatory legal requirement for installation of cookstoves in households/institutions of Rwanda. Step 2: Positive List Further, the PP distributing ICS free of cost to end users, hence there will be no revenue stream than sale of VCUs, hence the project activity instances will be automatically additional. The above can be checked with end user agreement and declaration from PP on the same. The PP has provided end user agreement and declaration to VVB to support free distribution of ICS.
6	Where a capacity limit applies to a project activity included in the project, no PAI shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows: Each PAI that exceeds one percent of the capacity limit shall be identified.	The GP does not have any specified capacity limits applicable, since there are no capacity limits defined within the applied methodology, VMR0006, ver. 1.1. Therefore, this eligibility criteria is not applicable.	N/A

	Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters. None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit.		
Eligibility conditions specific to inclusion of New PAIs			
7	The new PAI must occur within one of the designated geographic areas specified in the project description	The distribution of the ICS will be done in the jurisdictions of Rwanda. Each PAI distributes the ICS units to HHs in umudugudus (Kinyarwanda term used for villages) and cells which are distinct from each other. No two PAI or VCS Grouped projects in the host country Rwanda can have ICS units distributed to same HH. This is ensured by accessing the beneficiary list from the cell office,	1. Sample list of the beneficiaries that are to be provided the project ICS will be shared. 2. Project ICS distribution data, recorded during the HH visit. For PAI-1, records distribution (end user registration database) with clear delineation of project boundary i.e. host country of Rwanda is provided to the VVB for their reference.

		which is then verified on-site by the DelAgua team for any corrections in the number of beneficiaries. Post that, the final list of beneficiaries is reviewed by the umudugudu (village) chief. Distribution related information of ICS units specific to each PAI and other DelAgua grouped project activities is then recorded in separate databases using electronic data recording software, form.com. The unique identifiers of the distributed ICS unit reflect that it is distributed to a unique HH in the host country Rwanda.	
8	The new PAI must comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.	The eligibility criteria set must be complied by the PAIs at all points in time throughout the crediting period. If at any point, it is found that the PAI is not complying with the set of eligibility criteria, then it may be excluded from the crediting for the duration of the monitoring period, until it is able to demonstrate the compliance of the eligibility criteria.	The MR will act as the documentary compliance summary. However, for each specific eligibility criteria (as laid down above), the supporting evidence will be provided by the PP. The PAI-1 is part of VCS grouped project validation/verification, hence no further assessment.
9	The new PAI must be included in the monitoring report with sufficient	The new PAI will include all the technical, financial, geographic and other	The MR for the specific monitoring period will act as the supporting

	technical, financial, geographic and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	relevant information related to the compliance of the applicable set of eligibility criteria & also enable sampling by the VVB.	evidence for the stated condition/requirement. The PAI-1 is part of VCS grouped project validation/verification, hence no further assessment.
10	The new PAI must be validated at the time of verification against the applicable set of eligibility criteria.	The new PAI(s) shall be part of the monitoring, and therefore, will be validated against the criteria specified, by a VCS accredited/approved VVB.	The Verification Report based on the VCS template, shall be issued by the approved VVB. This will act as the supporting evidence against this requirement. The PAI-1 is part of VCS grouped project validation/verification, hence no further assessment.
11	The new PAI must have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	The PAI related evidence on the project ownership will be in the form of a distribution end user receipt poster. This will be signed by the end user, and a photograph of him/her holding the poster along with the ICS will be saved in the PAI database.	1. A photograph of the PAI distribution first end user holding the poster along with the ICS. 2. A scan of the poster, which will show the first distribution date. 3. The distribution database of the GP will provide the start date in the form of the first distribution made by the PAI. Waiver of carbon credits from the end users are submitted to the VVB for PAI-1.

12	The new PAI must have a start date that is the same as or later than the grouped project start date.	The new PAI will have a start date later than the GP start date.	This can be verified through the GP distribution database, wherein each new PAI can be verified for its start date. The PAI-1 is first instance and start date of grouped project activity is same as start date of PAI-1. The distribution record shared with VVB to support start date of PAI-1.
13	The new PAI must be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period (as set out in Section 3.4.4 of the VCS Standard, ver. 4.3 ⁴) and new instances are eligible for crediting from the start of the next verification period.	The PAI will only be eligible for crediting from the start of the verification period in which they were added to the grouped project.	1. The start date of the PAI can be checked through the distribution database. 2. The end date of the GP crediting shall be referenced from the VCS Project Registry webpage. 3. The overlap of monitoring periods may be checked through the previous monitoring reports (available publicly on VCS project registry webpage for those periods). The PAI-1, is first instance and start date of grouped project activity is same as start date pf PAI-1, hence no overlapping of crediting period.

⁴ The monitoring period of the monitoring report shall be a distinct time period that does not overlap with previous monitoring periods. Projects shall not be eligible for crediting of GHG emission reductions generated in previous monitoring periods. In addition, monitoring periods shall be contiguous with no time gaps between monitoring periods.

14	The new PAI shall not leave one VCS project and subsequently enroll in another VCS project.	It will be ensured that the PAI that is being enrolled under the VCS GP 3699 was not previously enrolled under any other VCS project, and will not be enrolled under other VCS project if it is removed from the VCS GP 3699.	This will be evidenced through a declaration signed by the PP for the PAIs that are enrolled under the VCS GP 3699. Since, for PAI-1 project proponent is the implementer, no further approval is required, however for the cases, where implementer other than PP, an approval on the same shall be given by PP in order to include the instance into this grouped project.
15	The new PAI shall adhere to clustering requirement that states the project proponent shall include in a singular project all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent	It will be ensured that PAI included within 10 kilometers of another instance of the same project activity by the same PP is in a singular project	PAI-1 involves distribution of improved cookstoves by same PP within 10 kilometers hence included in the same project. There are no other project instances by the same PP distributed within 10 kilometers for the same project activity.

1.5 Project Proponent

Organization name	DelAgua Trans Africa Stoves Pte. Ltd. (DATAS)
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Title	Mr.
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1.6 Other Entities Involved in the Project

Organization name	DelAgua Health Rwanda (Voluntary) Limited
Role in the project	
Contact person	James Beaumont
Title	Mr.
Address	c/o Cafferata & Co., Poinciana & West Mall Drove, Freeport, Grand Bahama, P.O. Box F-42614, Commonwealth of the Bahamas
Telephone	+250-790002992
Email	James.Beaumont@delagua.org

1.7 Ownership

DelAgua Trans Africa Stoves Pte. Ltd. (DATAS) is the owner of the VCS grouped project activity.

During the distribution of the improved cookstoves, the participating households sign an end user agreement between customer and DATAS. The end user agreement has customer information, unique identification number, product details etc. The agreement follows the requirements of para 3.7.1 of the VCS Standard ver4.4 which states the evidence to establish project ownership should be “An enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals which vests project ownership in the project proponent.”

The agreement confirms that the ownership rights of the VERRA project and the carbon assets generated from this project lie with the project proponent. The customer under the End User Agreement “releases all rights to the greenhouse gas reductions and carbon credits produced by the use of the clean energy product in the favour of DATAS and agree to not sell or transfer the GHG or carbon credits to any other third party or use these credits for any other purposes”

This will be evidenced through the below:

- Manufacturer agreements confirm that DATAS is the sole owner of all technologies purchased under this project. These agreements further confirm the manufacturer’s understanding that the carbon credits associated with use of the technologies shall be within the jurisdictions of DATAS.

- During the distribution of the ICS, the participating household sign an End User Agreement to confirm that the ownership rights of the carbon assets generated from this project lie with the project proponent.

1.8 Project Start Date

The start date for this Grouped Project is 04-November-2022, which is the first date from which actual distribution of ICS started.

1.9 Project Crediting Period

The project crediting period will be seven years, twice renewable, for a maximum total of 21 years.

The first crediting period will be 04-November-2022 to 03-November-2029, including both the dates.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	
Large project	☒

Year	Estimated GHG emission reductions or removals (tCO _{2e})
04-November-2022 – 31-December-2022	117,023
01-January-2023 – 31-December-2023	2,136,377
01-January-2024 – 31-December-2024	2,708,533
01-January-2025 – 31-December-2025	2,595,794
01-January-2026 – 31-December-2026	2,555,901
01-January-2027 – 31-December-2027	2,542,201
01-January-2028 – 31-December-2028	2,508,991
01-January-2029 – 03-November-2029	2,027,310
Total estimated ERs	17,192,130

Total number of crediting years	7
Average annual ERs	2,149,016

1.11 Description of the Project Activity

Distribution plan:

The distribution of ICS will occur at centralized distribution points, where households from surrounding villages will be educated on the proper use and maintenance of the technologies. Shortly following the initial distribution, CHWs will visit each household that received a technology to deliver the training again, ensuring that all members of the household are aware of proper use and maintenance.

The ICS distribution is planned in close coordination with the local government offices, e.g. District Offices, Sector Offices, and so forth. There is a structured plan through which the distribution activity is scheduled and then implemented on ground.

Project activity description (including the technologies or measures employed) and how it/they will achieve net GHG emission reductions or removals.

The project activity included in this project description is the distribution and installation of high efficiency ICS systems in PAIs within Rwanda.

The ICS units will be based on a “rocket stove” design which reduce GHG emissions from biomass burning through improved combustion efficiency of wood fuel and decreased wood fuel consumption. The project stoves improve heat transfer to cooking pots through the insulated walls and design of the stove top when compared to traditional cooking methods over open- and three-stone fires. The project stoves directly decrease wood fuel demand and reduce the pressure from wood cutting within the project boundary. A 30-50% saving of firewood when compared to three-stone fire cooking has been reported in certain models. The time spent foraging for wood fuel is thereby also reduced, saving time for especially women to engage in other economic driven activities.

ICS technologies planned under the GP for distribution:

The following ICS models are envisaged to be distributed in the initial PAIs –

Model-1:

Name of the stove – Kuniokoa Generation 2

Stove manufacturer – Burn

Stove Type – Natural Draft

Stove Body – Stainless Steel

Stove height – 32.1 cm

Stove diameter – 28.2 cm

Thermal efficiency with pot skirt– 51.3%

Thermal efficiency without pot skirt – 43.4%

High Power Output – 2.9 KW

Stove lifetime – 10 years

Figure 1: Kuniokoa Stove Model reference picture



Model-2:

Name of the stove – SSM S32-X⁵

Stove manufacturer – SSM

Stove Type – Natural draft

Stove Body – Stainless Steel

Stove height – 28 cm

Stove diameter – 32 cm

Thermal efficiency with Pot Skirt – 50.1%

Thermal efficiency without Pot Skirt – 45.7%

Stove lifetime – 10 years

Figure 2: S32-X Stove Model reference picture

⁵ <https://www.ssmcook.com/product/wood-stove-ssm32-x/>



List of main manufacturing/production technologies arrangements, systems and equipment involved.

Detailed application and information on the ICS technology that is going to be distributed under the new PAIs will be described in the upcoming MRs, elaborating the information under relevant section of the MR for the newly included PAIs. PP will source the best available technologies for distribution under the GP. The project already distributed 130,633 BURN Kuniokoa at the time of registration for this project. The ICS model selection at PAI level will be based on key attributes such as thermal efficiency, durability, cultural acceptability, and manufacturing quality and warranty. From time to time, requirement gathering will be done on a continuous basis, seeking comments from the end users (through monitoring surveys, feedback requests on call center, feedback via register available at different sites, and anonymous feedback through CHWs), and based on these user feedbacks, newer models will be introduced to the GP, and new PAIs will be designed to accommodate for their variability in terms of design, specifications, and geographies within the host country (wherever required).

It will be verified by the PP that the project technologies are, at a minimum, meeting the technical specifications and requirements set in the respective methodologies (under the eligibility criteria).

At present, DelAgua is sourcing the ICS from different manufacturers, deploying them at its central warehouse through imports, and then distributing them in a scheduled manner. Each distribution involves various steps, which are elaborated in the relevant section of this document. In this whole process, quality of the equipment being distributed will be checked on spot basis. Each lot of the distribution batch will undergo spot checks, as it is an integral part of Quality Assurance at DelAgua. The ICS units that will be found as faulty, will not be distributed to the end users. At present, only rocket stove models are planned for the upcoming distribution PAIs, as their distribution is found both beneficial to the locals (usage and maintenance) as well as a successful one from the implementation point.

The types and levels of service

Type of Service: The ICS technology will deliver a specified high-power thermal efficiency of at least 25% and evidenced by a manufacturer's specification and WBT certificate. This is a significant improvement on the baseline open fire thermal efficiency default of 10%.

Under the baseline scenario, households use non-renewable woody biomass for their fuel needs for cooking. The baseline appliance being replaced by the project is an open fire (typically based on the three-stone concept).

For each PAI, the project boundary will be defined as being within the geographic borders of Rwanda. The project boundary for each Project Activity Instance is the physical, geographical location of the ICS distributed under it within the geographical boundaries of Rwanda.

The map displays the administrative districts of Uganda, color-coded and labeled: Northern (light green), Western (light green), Central (light green), Eastern (light green), Southern (orange), and Kigezi (pink). Major cities like Kampala, Jinja, and Mbarara are marked. The map also shows Lake Kyoga, Lake Kyoga, and Lake Kyoga. Neighboring countries (Uganda, Tanzania, Burundi, DR Congo) and a legend for district types are included.

The baseline scenario is the same as the conditions existing prior to the Grouped Project initiation.

For further information regarding this, please refer Section 3.4 (Baseline Scenario) of this document.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The GP complies with applicable local laws and regulatory frameworks. Under Rwandan law and regulation, none of the activities or processes relating to the manufacture or distribution of Improved cookstoves require Environmental Impact Assessments not meet any of the screening criteria in Appendix 2, page 40 of the EIA guidelines⁶. The VCS grouped project is a regulatory surplus and is not mandated by any law within the host country. Further information on this may be accessed from the Section 3.5 within this document.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

No. This proposed VCS Grouped Project is neither registered nor is seeking registration under any other GHG programs.

1.15.2 Projects Rejected by Other GHG Programs

Not applicable, as this project is a new Grouped Project activity and has not been rejected by other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The grouped project does not reduce GHG emissions from activities that are included in any emissions trading program..

1.16.2 Other Forms of Environmental Credit

The VCS GP 3699 has not sought or received another form of GHG-related environmental credit, including renewable energy certificates.

1.16.3 Supply Chain (Scope 3) Emissions

Not applicable

⁶ General Guidelines and Procedure for Environmental Impact Assessment, Republic of Rwanda, November 2006 (See filename: RW_EIA_Guidelines_Final_versionI_Nov_2006)

As per VCS Standard version 4.4, this section (Scope 3 emissions inventories) is effective 1st January 2024 onwards hence this section is not applicable for the current monitoring period i.e., 04-November-2022 to 31-December-2022.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The GP is contributing towards sustainability, which is also verifiable through the SDG indicators selected for the specific activities that are going to be monitored for their fulfilment.

Following sustainable contributions were part of the PAI – 1 implementation (Distribution of ICS in Rwanda):

SDG-1: No Poverty

Approach: For arriving at the indicator value, we monitor the relevant parameter for this SDG. In this case, the questions specific to the access to cooking technologies that are available to the households (HHs) prior to the project ICS distribution is counted towards the estimation of the indicator to this SDG.

Relevant SDG Indicator: 1.4.1 Proportion of population living in households with access to basic services

Available Data: The parameter that was monitored to calculate the number of HHs living without access to basic services was the number of beneficiary HHs. This is because the number of HHs that are unable to purchase any improved cooking devices can be considered as the ones living without access to the basic services such as clean cooking. The list of HHs which are to be distributed with the project ICS is provided by the government of Rwanda, and is based on the economic classification list of the population of Rwanda.

Monitoring Requirement(s): The new project ICS beneficiary/user will be considered as living without the access to clean cooking, and the number of people using the project ICS as their primary device will be considered towards the estimation of this SDG impact. Clean cooking is considered as a basic service, and access to it is an addition to the SDG impact.

SDG-3: Good Health and Well Being:

Approach: For arriving at the indicator value, we monitor the relevant parameter for this SDG, specifically the SDG indicator. This SDG relates to health, and DelAgua's project ICS is improving health for every individual who is receiving it.

Relevant SDG Indicator: 3.9.1 Mortality rate attributed to household and ambient air pollution

Available Data: Regarding indicator 3.9.1, at every project ICS beneficiary HHs, ones who are experiencing any improvement (or no improvement or deterioration) in their ambient air quality as compared to pre-project scenario will be considered as an input.

Monitoring Requirement(s): The data was collected at the time of monitoring survey. Beneficiary HHs for VCS GP will be sampled for monitoring as per monitoring plan, and these are then requested to respond to survey questions, which is then recorded within the monitoring survey database. The percentage of people experiencing an improvement in air quality (i.e. decrease in smoke levels in HH due to cooking) will count towards a reduction in the mortality rate attributed to household and ambient air pollution. The number of people will be equal to the product of average household size identified during monitoring surveys and the number of total HHs received the ICS and the proportion of HHs which respond that they've experienced a decrease in smoke levels from the baseline (in response to the survey question).

SDG-4: Quality Education

Approach: Here, each household is trained for sustainable practices, best usage practices for ICS usage, and utilization of fuel resources. Apart from this, the project being one of its kind, provides education and training to 5000+ Community Health Workers (CHWs) engaged with us for distribution and end-user engagement.

Relevant SDG Indicator: 4.7.1 Extent to which (i) global citizenship education and (ii) education for sustainable development are mainstreamed in (a) national education policies; (b) curricula; (c) teacher education; and (d) student assessment

Available Data: (i) Number of project ICS beneficiary HHs who were visited by DelAgua staff for HH visit surveys are used to calculate the parameter. This data is continuously recorded in the distribution master database.

(ii) Number of CHWs that were trained for the project activity implementation (distribution and surveys).

Monitoring Requirement(s): The distribution master database and the CHW training data is used to calculate the number of people trained under the project activity.

SDG-5: Gender Equality

Approach: In this case, the number of ICS beneficiary HHs will be questioned over their time spent for collecting fuel (firewood).

Relevant SDG Indicator: 5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location

Available Data: Monitoring Survey for VCS GP includes question on time spent on collection of fuel.

Monitoring Requirement(s): The value is recorded in the monitoring surveys, and then saved in the survey database. The total number of HHs which acknowledge that the amount of time spent on collection of firewood has changed will be arrived by calculation of survey data – i.e. multiplying the proportion of survey respondents accepting a decrease in time spent on cooking fuels with the total number of households.

SDG-7: Affordable and Clean Energy

Approach: For arriving at the indicator value, we monitor the relevant parameter for this SDG, specifically the SDG indicator.

Relevant SDG Indicator: 7.1.2 Proportion of population with primary reliance on clean fuels and technology

Available Data: Distribution Master Database includes the data for this, where each HH where project ICS is distributed, acts as an input. The data within the database is updated on a continuous basis.

Monitoring Requirement(s): Distribution Database is exported for the monitoring period, and then the number of beneficiary HHs are multiplied with the average HH size of Rwanda (which is 4.3 in 2021). This gives the number of people who are now benefiting from the clean and affordable energy for the Monitoring Period.

SDG-8: Decent Work and Economic Growth

Approach: For arriving at the indicator value, we monitor the relevant parameter for this SDG, specifically the SDG indicator.

Relevant SDG Indicator: 8.3.1 Proportion of informal employment in total employment, by sector and sex

Available Data: The salary slips for the CHWs act as the data source since these are not on DelAgua's payrolls. However, there are different data sources (number of mobile phones assigned to individuals during distribution, number of CHWs trained for a specific distribution event), which are used to total the CHWs working for DelAgua.

Monitoring Requirement(s): This data is continuously collected and is obtained from Training and Account teams.

SDG-13: Climate Action

Approach: For this SDG, there are no separate monitoring requirements. We declare the GHG Emission Reductions achieved by project in the monitoring period. This is verified by a VCS accredited VVB, and the resultant VCUs are then issued, acting as the net impact contributed towards this SDG.

Relevant SDG Indicator: Not applicable.

Available Data: This can be accessed from the Monitoring Reports that are publicly available.

Monitoring Requirement(s): All the monitoring requirements are already detailed in the Monitoring Reports and corresponding ER Sheets.

SDG-15: Life on Land

Approach: For arriving at the indicator value, we monitor the relevant parameter for this SDG, specifically the SDG indicator.

Relevant SDG Indicator: 15.4.2 Mountain Green Cover Index

Available Data: (i) Baseline (pre-project) survey data, (ii) Monitoring Surveys (periodic), (ii) Registered PD&MR and ER Sheet

Monitoring Requirement(s): This parameter is fixed for per ICS (HH level) firewood consumption, which is then multiplied with monitored values (such as usage rate), other relevant values as per the equations provided in the methodology applied for VCS GP. Based on the calculations, per HH level firewood savings are calculated.

This is then multiplied with the total number of ICS distributed. It gives the total amount of firewood saved by the PAI (or GP) in the specified monitoring period. This will be the amount of firewood that is saved from being harvested from the valuable mountain forest ecosystem.

The SDG impacts of the distribution program and monitoring related discussion is provided in the below sections. In addition to that, the VCS GP 3699 will seek registration under SD VISTA for the labeling of its additional benefits that relate to UN SDGs.

1.17.2 Sustainable Development Contributions Activity Monitoring

The SDG impacts anticipated from the GP, and the plan for monitoring those impacts is described in the table below.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	1.4	1.4.1 Proportion of population living in households with access to basic services	Implemented activities to increase	Cooking is considered a basic service, as food is considered a basic need of humans and other forms of life. Therefore, improved cooking access is a basic need as well. For PAI-1, the number of HHs receiving the project ICS now have access to basic cooking services, since all of the ICS end users surveyed mentioned that project ICS is their primary cooking device. Therefore, 100% of the users surveyed are being considered as the number of people with access to basic services. This number stands at 130,633 (i.e. the number of ICS units distributed in M.P.-1) for the first monitoring period. So, a total of 130,633 households now have access to basic cooking services when compared with the baseline.	This is the first verification for this project activity, so the current contributions are equivalent to the contributions over the project lifetime.

2)	3.9	3.9.1: Mortality rate attributed to household and ambient air pollution	Implemented activities to decrease	Mortality rate attributed to HH and ambient pollution will decrease for those HHs using the DelAgua improved cookstove. The improved design specifications of the project ICS model(s) ensure there is no pollution in the households. Out of the total population of Rwanda, a total of 561,834 people now have less risk towards respiratory ailments due to smoke inside their homes due to the implementation of VCS 3699 (for first monitoring period).	This is the first verification for this project activity, so the current contributions are equivalent to the contributions over the project lifetime.
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3)	4.7	4.7.1: Extent to which (i) global citizenship education and (ii) education for sustainable development are mainstreamed in (a) national education policies; (b) curricula; (c) teacher education; and (d) student assessment	Implemented activities to increase	The project activity educates and trains the team involved in the distribution of ICS, and also educates the end user HHs about the sustainable cooking practices. With this, it can be said that the local population is learning sustainable practices and healthy lifestyle without spending anything. i) The number of CHWs that were educated (trained for distribution, 6-monthly household visits, surveys, and training on sustainable cooking practices to be disseminated to end user households) by the Grouped Project (for PAI-1) by the end of M.P.-1 are 4,783. ii) The number of end users per household that are educated for the stove usage and apply sustainable cooking practices are equal to the exact number of households where project ICS are distributed. PAI-1: 130,633 households educated.	This is the first verification for this project activity, so the current contributions are equivalent to the contributions over the project lifetime.
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4)	5.4	5.4.1: Proportion of time spent on unpaid domestic and care work, by sex, age and location	Implemented activities to decrease	The % of HHs where time spent on collecting fuel and cooking meals has decreased will increase with the implementation of this project activity. During monitoring surveys, it was identified that 97.5% of respondents acknowledge that they experienced a decrease in the time spent on collection of cooking fuel (firewood). This, when multiplied with the total number of end user HHs, give 127,367 (rounded down to nearest whole number). So, 127,367 HHs have women spending less time on unpaid domestic work (i.e. collection of firewood for cooking).	This is the first verification for this project activity, so the current contributions are equivalent to the contributions over the project lifetime.
5)	7.1	7.1.2: Proportion of population with primary reliance on clean fuels and technology	Implemented activities to increase	The number of HHs with access to clean cooking will increase with the project activity implementation. For M.P.-1, PAI-1 ensured that 130,633 HHs now have primary reliance on clean cooking technologies.	This is the first verification for this project activity, so the current contributions are equivalent to the contributions over the project lifetime.

6)	8.3	8.3.1: Proportion of informal employment in total employment, by sector and sex	Implemented activities to increase	The project activity has a database of CHWs working for it. This database is used to provide the actual number of CHWs working for the project activity. The number of permanent employees working on the DelAgua payroll will not act as a part of this indicator since it is only focused on the informal employment. However, the number of estimated people who worked in the manufacturing of these ICS, and those working for the delivery and transportation of these ICS will be considered towards the indicator. Even if the PP is not directly employing the ironsmiths or technicians working on contract at the manufacturer's facility, purchasing of these stoves will become an indirect source of their income. During M.P.-1, the project activity provided informal employment to around 6000 people. This is explained below: 4783 CHWs at DelAgua, 900+ full time employees working with the ICS manufacturer & suppliers (Burn ICS⁷), 200+ different transportation rounds/people worked for transporting the stoves from Kenya to Rwanda (Sample transportation invoice shared). This totals to around 6000 people.	This is the first verification for this project activity, so the current contributions are equivalent to the contributions over the project lifetime.
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7)	13	13: Take urgent action to combat climate change and its impacts	Implemented activities to increase	The emission reductions units will be generated by this project activity, thus mitigating Climate Change. For the first monitoring period, the number of VCUs calculated for PAI-1 are 59,473 tCO_{2e}.	This is the first verification for this project activity, so the current contributions are equivalent to the contributions over the project lifetime.
8)	15	15.4.2: Mountain Green Cover Index	Implemented activities to increase	The implemented project activity will lead to a decrease in the deforestation in the mountainous forest ecosystem. This eventually will lead to an increase in the mountain green cover index compared with the baseline. From the ER calculations made for PAI-1, 3.26 tons per project ICS is the calculated value of firewood savings. If we multiply this with the number of HHS, we'll get the amount of firewood saved by the GP, which is 426,568 tons for M.P.-1.	This is the first verification for this project activity, so the current contributions are equivalent to the contributions over the project lifetime.

⁷ <https://www.linkedin.com/company/burn-co/?originalSubdomain=ke>

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage as per applied methodology VMR0006 V1.1.

Commercially Sensitive Information

All information in this Grouped Project Description is public.

Further Information

Not Applicable.

2 SAFEGUARDS

2.1 No Net Harm

The project undertakes the distribution of improved cook stoves. As per the Laws of Rwanda, the project undertaking such activities does not require any Environmental Impact assessment. This is as per Ministerial Order No. 004/2008: Ministerial Order establishing the list of works, activities and projects that must undertake an environment impact assessment.⁸ The fact that this project activity does not need environmental assessments, supports that it does not cause environmental harm.

The project, on the contrary, shall have several positive social and environmental benefits due to its implementation. The project will lead to improved indoor air quality, lower air emissions, and also positive health impact.

2.2 Local Stakeholder Consultation

The procedures or methods used for engaging local stakeholders

The Local Stakeholder meeting, under VCS Standard v4.4 with NGOs, Academia, other project developers, and any other entity who concerns itself with the DelAgua clean cookstove distribution was conducted at national level on 01-March-2023.

⁸ <https://www.elaw.org/rwanda-environmental-impact-assessment-law-and-regulations>

The relevant stakeholders for the LSC were invited using various mode to get the participation/feedback which included publication in newspaper “The New Times” dated 03 & 13 February 2023, email sent to identified NGOs and other organizations dated 04 & 13 February 2023 and advertisement in radio channels. The local stakeholder consultation meeting was held on 01-March-2023 from 9.00 AM to 11.00 AM at Gorillas Golf Hotel, Nyarutarama, Rwanda.



Outcome of the local stakeholder consultation

The agenda of the meeting was to showcase the VCS grouped project being implemented and invite local stakeholders to understand about the project, submit their grievances, and raise their concerns, wherever required. Below are the steps followed during stakeholder consultation meeting

1. The meeting started with presentation on grouped project activity and associated benefits to society, ecology and economic aspect due to use of improved cookstoves.
2. The sample cookstove units, which were being distributed under VCS 3699 implementation, was demonstrated to the local stakeholders present for its usage.
3. During the meeting, stakeholders were invited to raise their queries and concerns. These were duly responded to during the meeting itself.



Stakeholder Feedback

Below are the queries raised by the attendees

Q.1. Whether current grouped project activity of DelAgua involves only improved cookstove distribution?

Ans: The project activity involves only Improved cookstove distribution.

Q.2. Which districts/region the project is planned under the current project activity?

Ans: The PP representative explained that the project boundary for the grouped project activity is host country Rwanda and various districts will targeted where ICS penetration is still less or not there.

Q.3. Whether any changes in ICS models to be used under current project than ICS distributed under other ongoing projects as DelAgua group is already having other registered projects and distributing ICS from past many years.

Ans: The technology upgrades with time and PP uses the state-of-the-art ICS technology available at the time of start of the grouped project activity, under current project the PP is distributing Burn make Kuniokoa generation 2 and S32 models which are having more than 50% thermal efficiency.

There were no negative comment/concern raised by stakeholders attended the meeting.

The LSC meetings were recorded using digital methods (audio-visual, photographic, and pdf records of the Q&A along with attendance), and paper format of the meeting forms. The invites

are also shared with the local stakeholders a month before the meeting date. The evidences are shared with the VVB for their validation.

Additional Stakeholder Engagements

In addition, the PP conducted numerous rural stakeholder engagements prior to the project cookstove distributions at a District and Sector level in July 2022. These meetings were intended to do the following for the rural communities who would be the ultimate beneficiaries of the stoves:

1. To gather local people together and inform them about the VCS grouped project activity.
2. To make them aware about the Carbon Finance without which this VCS grouped project could not be established.
3. To make the local people understand the benefits of clean cooking.
4. To demonstrate the working of sample cookstove that is aimed for the distribution.
5. To collect their feedback, queries, concerns, and respond those at the meeting itself.
6. Provide updates to community members of when the stoves will be arriving ready for disbursement and education.

These rural engagements were not in line with para 3.18.3 of VCS Standard 4.4 hence PP conducted local stakeholder consultation in line with the requirements on 01-March-2023.

Ongoing communication with the Local Stakeholders

As far as continuous grievance of the local stakeholders is concerned, the required measures are already in place to account for a grievance through any media.

Following are the channels through which the local stakeholder can communicate with the PP during the implementation of GP:

1. PP (DelAgua) has a toll-free helpline for the relevant stakeholders to connect with them in case they require any assistance.
2. The DelAgua CHW network is always ready to obtain any feedback from the end users and local authorities. Most of the feedback is received by PP through this medium, since a lot of people in rural Rwanda do not have access to phones.
3. PP distribution team informs the end users during distribution that they may write their feedback on paper, and submit it to the nearest cell office, and PP team will receive it through the government officials.

These medium of interaction with local stakeholders will be verified by the VVB at the time of validation, and shall remain open for Verra for review.

In line with the VCS requirements, these documents shall be recorded and maintained for a minimum duration of 2 years post the end of the project implementation.

During the current monitoring period, there were no grievance or concern raised by any stakeholder.

2.3 Environmental Impact

The project undertakes the distribution of improved cook stoves. As per the Laws of Rwanda, the project undertaking such activities does not require any Environmental Impact assessment. This is as per Ministerial Order No. 004/2008: Ministerial Order establishing the list of works, activities and projects that have to undertake an environment impact assessment⁹.

2.4 Public Comments

The public commenting period (10 November - 10 December 2022) is over for the GP, and no comments have been received during the public commenting period.

2.5 AFOLU-Specific Safeguards

Not applicable since the GP is not an AFOLU activity.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The Grouped Project activity will involve PAIs that will apply the following methodology, approved under VERRA:

- VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves – version 1.1

In the context of applied methodology, the associated tools and guideline documents applicable for the Grouped Project include:

- CDM TOOL30 “Calculation of the fraction of non-renewable biomass” Version 04;

⁹ <https://www.elaw.org/rwanda-environmental-impact-assessment-law-and-regulations>;

Specific URL to the Order No. 004/2008:

<https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Felaw.org%2Fsystem%2Ffiles%2Fattachments%2Fpublicresource%2FMINISTERIAL%2520ORDER%2520N%25C2%25B0004%2520ESTABLISHING%2520THE%2520LIST%2520OF%2520WORKS.docx&wdOrigin=BROWSELINK>

- CDM Guideline “Sampling and surveys of CDM project activities and programmes of activities” version 04;
- CDM Standard “Sampling and surveys for CDM project activities and programmes of activities” version 09.

3.2 Applicability of Methodology

The applicability criteria of the methodologies and tools used are justified at the project activity level. This is reflected in the table below.

Table 5: Fulfilment of VMR0006, ver. 1.1 Applicability criteria for PAIs to be included

S. No.	Applicability Condition	Justification of applicability
1	Project activities shall be implemented in domestic premises or in community-based kitchens.	Since the PAI distributes technology that is appropriate for domestic use to household end users i.e. improved cookstoves for cooking, thus this meets the applicability
2	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer’s specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	ICS distributed in each PAI has a thermal efficiency of at least 25% Baseline and project fuel for each PAI has been recorded to ensure that wood fuel is the sole source of energy.
3	Both ‘Projects’ and ‘Large Projects’ can use the methodology	As per the VCS requirements, the VCS GP 3699 is classified under the Large Project category. As such, PAIs also fall under the Large Project category.
4	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	Statistics from the Forest department show that forests were estimated to cover 659,000 ha in 1960, which has reduced to 240,746 ha in 2007 ¹⁰ . This reflects a loss of approximately 64 per cent of forests between 1960 and 2007, which is more than 1.3 per cent per year. Firewood

¹⁰ <https://www.rema.gov.rw/soe/chap6.pdf>

		remains the main source of domestic energy for more than 90 per cent of Rwandan households, s. The above can also be confirmed from Forest Resource Assessment report 2020 for Rwanda, which states around 45.27% loss in forest from 1984 to 2015¹¹. Further the study conducted in June 2009 in various districts of Rwanda reveals that 98% of rural households use wood fuels as their main source of energy for cooking, which consists of 92% firewood, 5% charcoal and 3% other residue¹². A recent study conducted by Ministry of Land and Forestry, Rwanda in 2017¹³, reveals that wood demand and supply ratio is 2:1 and the shortage is projected to increase in the future unless alternative sources of wood energy are sought. . The Business as Usual scenario on wood supply/demand, estimates the deficit between wood supply and demand to be 4.3 million tonnes (oven dry weight) in 2017, which is projected to increase to 7.5million tonnes by 2026. This is due to a high increase demand for fire wood and wood for charcoal. This must imply over-exploitation of already low stocked forests. The analysis also shows that 93% of total wood demand mentioned above is for cooking purpose (57% firewood and 36% charcoal). It can therefore be concluded that non-renewable biomass has been used in Rwanda since long back before 31st December 1989.
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¹¹ *FRA 2020 Étude de bureau, Rwanda*

¹² <https://www.cleancookingalliance.org/binary-data/RESOURCE/file/000/000/36-1.pdf>

¹³ https://www.climateinvestmentfunds.org/sites/cif_enc/files/fip_final_rwanda.pdf

5	For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	Biomass residues are not processed as a fuel to be used in the GP. The PAI uses non-renewable biomass (firewood) as source of fuel for the improved cookstove. No processed biomass residue is used.
6	The Project Design Document shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo)	In order to avoid double counting of emission reductions through the method of distribution, each cookstove has a unique identifier number and the distribution of each stove is logged onto a database which includes unique information for each recipient Each PAI has a system for avoiding double counting with data being collected via a smartphone app (where possible), or via paper forms, including: 1. Date of distribution 2. A unique serial number on each technology 3. Available end-user data: name, phone number, address/physical location, ID number
7	The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid	The End User Agreement confirms that the ownership rights of the ER/Vcus generated by the PAI reside with the PP. The signature of each end user confirming

	that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	their understanding of this agreement has been collected at the point of distribution. The End User Agreement is in English and translated into a local language where required. If necessary, it is read out to the End User. All rights to distribute the delivered stoves and claim the resultant emissions reductions belong to the PP. The GP doesn't include any PAI that has been rejected by another GP, or PA-DDs of any other GHG programs.
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Tool 30 - Calculation of the fraction of non-renewable biomass

Applicability criterion	How the project complies
This tool may be used by: (a) DNAs to submit region- or country-specific default f_{NRB} values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or (b) project participants to calculate project- or PoA-specific f_{NRB} values.	Option (b) applied to calculate the project specific f_{NRB} value as described under in the separate f_{NRB} spreadsheet submitted to VVB.
Project participants and DNAs shall identify and clearly delineate the applicable area for which f_{NRB} is determined. For project- or PoA-specific f_{NRB} values, project participants shall assess and use the area from which woody biomass is sourced for end-users included in the project activity and justify the selection of the area in CDM project design documents. For region- or country-specific f_{NRB} values, DNAs shall specify the applicable area.	The PP has chosen host country Rwanda as region for the project activity.

Project participants and DNAs may choose between the following two options to determine the value of f_{NRB}: (a) Use the default value as provided in TOOL33; or (b) Calculate f_{NRB} by determining the share of renewable and non-renewable woody biomass in the total quantity of woody biomass consumption for the country/region or the project area (hereinafter referred as the applicable area) following the procedure and requirements in the paragraphs below. The project participants shall compare and analyse the calculated values against the values for f_{NRB} reported in relevant scientific literature and justify any differences. This analysis shall be included in the appropriate section of the PDD. The relevant scientific literature should include at least: (i) Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional woodfuels. Nature Climate Change, 5(3), pp. 266–272.	The PP has chosen option b), wherein a third-party consultant has established the f_{NRB} for host country. Please refer Appendix-1 for calculation details and comparison with literature.
If the f_{NRB} value is estimated at the national level, as a cross check, project proponent shall compare the value of estimated NRB with the product of: i) total average above ground biomass tonnage of the area of forest areas deforested in recent past (tonnes/ha), and ii) most recent available observed annual rate of deforestation (ha/yr). If the estimated NRB value is more than 10% above the value calculated as per the product of biomass and deforestation rate, justification shall be provided for the higher value for NRB	Please refer Appendix-1 for cross check.

3.3 Project Boundary

The Grouped Project boundary is defined as per the applied methodology VMR-0006, ver. 1.1.

Table 7: The GHG sources & sinks within the project boundary



Here, the change in mass of firewood between baseline and project scenario is $m_1 > m_2$.

The change in energy consumed for cooking between baseline and project scenario is $E_1 < E_2$.

The systems that were applied for the distribution activity to take place, are described in the Section 5.3 of this document.

3.4 Baseline Scenario

The baseline scenario is the continued use of traditional cookstove such as three stone-fired cookstove using non-renewable wood fuel (firewood) by the target population in rural and semi urban areas to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. According to the applied methodology on page 19 “A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney”. The baseline three-stone cookstove has an efficiency of 10%¹⁴.

PP conducted baseline survey and baseline KPTs between September to December 2021, to assess the type & quantity of fuels consumed and the cooking technologies in use in pre-project scenario. The PP initially conducted a desk-research to identify regions (districts) where use of traditional cookstoves and wood fuel is predominant. This was followed by designing a survey

¹⁴ <https://verra.org/wp-content/uploads/imported/methodologies/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf>

questionnaire to validate the baseline scenario in the selected regions. The baseline scenario was identified as the continued use of traditional cookstove (e.g. Rondereza) without chimney or grate or three-stone fired for preparation of meal. More than 90% households in the rural and semi-urban areas are relying on firewood for cooking.

Apart from the above, there are several documentary evidences in the form of recently published reports and papers that can support the fact that the project's baseline was usage of firewood and unimproved cooking devices.

Some of these are listed below:

1. Ministry of Infrastructure: Biomass accounts for 85% of all energy consumed¹⁵. Under the National strategy for transformation, the sector objective is to halve the number of households using traditional cooking technologies to achieve a sustainable balance between supply and demand of biomass through promotion of most energy efficient technologies.
2. Rwanda Integrated Household Living Conditions Survey, EICV5: In rural areas, firewood remains the most common type of cooking fuel, used by 93% of the households¹⁶.
3. Rwanda first Biennial Report to UNFCCC: Biomass (mostly wood fuel), is the most used fuel in households for cooking (83 % of households in 2019) and as a source of energy (86.3% in 2016/2017)¹⁷.

PP has conducted the baseline surveys (KPTs) in line with the applied methodology VMR0006 ver 1.1 and KPT protocol. PP used guidance provided in para 8.4-b of VMR0006 version 1.1. to conduct the baseline survey. The details of the sample size calculation for baseline surveys, sample selection criteria, procedures used for conducting KPT, and the results of KPT are as follows:

Sample Size Calculation:

The sample size was determined using guidance provided in para 8.4-b of VMR0006 version 1.1, which specifies a minimum sample size of 100 for a population greater than 1,000. PP targeted a sample size of 150 which meets the required criteria. In order to ensure that this approach was most conservative and satisfied all thresholds the PP also quality assured against the calculation of Bold as a mean-based parameter.

PP therefore took the following approaches:

Firstly, the sample size for the baseline KPTs was calculated in line with section 8.4(b) of the applied methodology, using the CDM Guideline for Sampling and Surveys, version 4.0. Using CDM Guidelines, the sample size for parameter Bold (mean parameter) was 6.

¹⁵ <https://www.mininfra.gov.rw/digital-transformation-1-1>

¹⁶ <https://www.statistics.gov.rw/publication/eicv-5-main-indicators-report-201617>

¹⁷ https://unfccc.int/sites/default/files/resource/Rwanda%20First%20Biennial%20Update%20Report_Final_V.pdf

Secondly, PP compared the sample sizes with the alternate approach provided by VMR0006 version 1.1 in section 8.4(b), which specifies a minimum sample size of 100 for a population greater than 1,000.

Thus, by utilizing both approaches, the PP ensured that calculated sample size met all appropriate methodological thresholds and is representative of the entire target population. PP then oversampled to account for potential non-responses, data quality concerns, and hardware errors. This process resulted in a total target sample size of 150.

Sample Selection Criteria:

Rwanda has a total of 5 provinces – Eastern, Western, Northern, Southern, and Kigali. However, the 5th province, i.e. Kigali¹⁸, includes 100% urban/peri-urban households and therefore is not included in the sampling procedure as these households lay outside the scope of the project.

The targeted sample size of 150 was divided among the four provinces of Rwanda - Eastern, Northern, Southern, and Western - resulting in approximately 35 samples per province. PP covered 16 out of 27 rural districts across these four provinces of Rwanda as shown below.

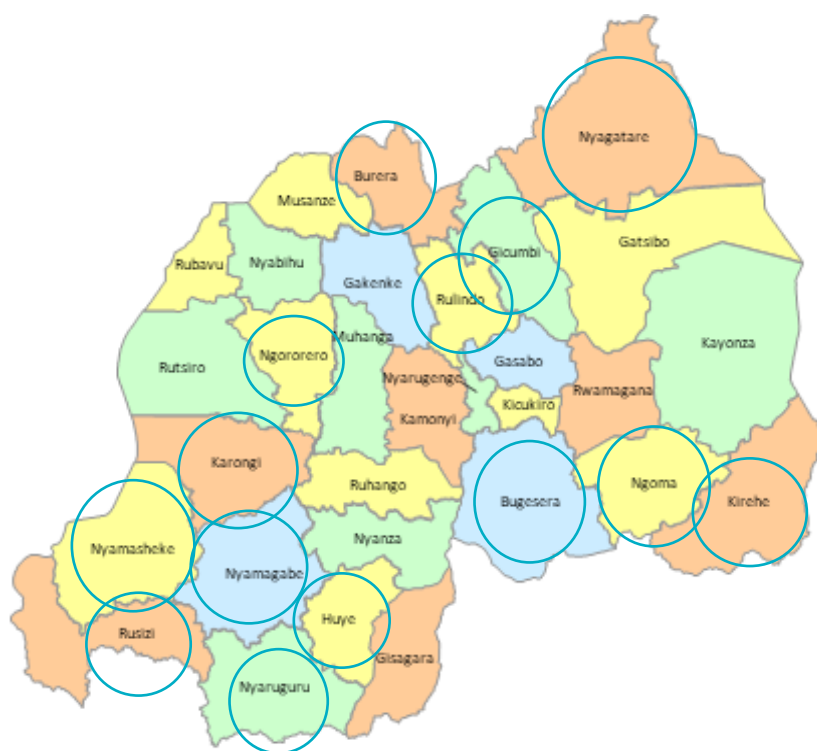


Figure 1: District Covered for Baseline KPTs

Once a target of 35-40 samples per province was confirmed by the technical team, the operational team, who has received KPT data collection training from Berkeley Air Monitoring

¹⁸ Administrative Structure of Govt. of Rwanda

Group (BAMG), was instructed to target 7-8 villages per province and 5 samples per village. As per the literature review¹⁹²⁰²¹, cooking behaviour in the entire country is quite similar. Major dishes being cooked are beans, stew, rice, vegetable stew, porridge, leafy vegetable stew, Irish potatoes, and green banana. Considering similar cooking practices across the country, the selection of these villages was left up to the field team to account for logistical challenges throughout rural Rwanda. This was on the condition that the villages selected did not have any criteria that would make them non-generalizable to the wider population, and they were culturally, and socio-economically homogenous with the rest of the province. To identify households, a random walk was taken, where every 5th household, when walking in a random direction, was approached, and assessed against the protocol criteria. If they met them, they were invited to take part in the KPT.

Out of the 150 target samples, PP successfully conducted 137 KPTs in line with the KPT protocol. The sample coverage is as follows:

Province	Sample Covered
Eastern	35
Northern	34
Southern	32
Western	36
Total	137

Procedures followed for Conducting KPT:

The KPT surveys were conducted by a competent team trained for the KPT surveys by Berkeley Air Monitoring Group (BAMG) for performing KPTs, based on the KPT protocol. Once the villages were selected by the operational team, enumerators were issued with protocols for the selection of the individual households. These protocols outlined that the selected households should, but are not limited to, the following criteria in accordance with the KPT protocol:

- Have the person who is primarily responsible for cooking available for visits once a day for the next 4 days;
- Not have any events scheduled that would disrupt normal cooking practices during this time (e.g Weddings, community events);
- Primarily cook on wood

Additionally, the enumerators were instructed to follow, but are not limited to, the following criteria in accordance with the KPT protocol:

¹⁹ <https://mecs.org.uk/wp-content/uploads/2022/04/Cooking-Diary-Study-Rwanda.pdf>, table 6, page 20

²⁰ <https://mecs.org.uk/wp-content/uploads/2022/02/MECS-EnDev-Rwanda-eCooking-Market-Assessment.pdf> - page 8-9

²¹ National Survey on Cooking Fuel Energy and Technologies in Households, Commercial and Public Institutions in Rwanda, Ministry of Infrastructure and Ministry of Finance, December 2020.

- Please avoid a measurement campaign during special occasions, holidays, or any day when cooking is not normal and be aware of local events like market days that may involve above-average fuel consumption. Be aware that 3 days of testing involves 4 days in contact with the family—the first day is spent briefing families, as explained in the next step.
- To explain the purpose of the test to the family members and arrange to measure their fuel consumption.
- Stress to household members that their cooking practices should remain as close to normal as possible for the duration of the test.
- Explain to family members that the surveyor will visit the respondent daily (for the next ~3 days) to collect the information.
- Ask the family to define an inventory area to store the fuel during the test. Record the weight and moisture content of the initial stock of fuels and if the family is going to collect or purchase fuel during the days of the test, ask them to keep newly collected or purchased fuel separate from fuel that has already been tested for moisture and weighed.
- Visit each household at roughly the same time each day, without being intrusive.
- Record fuel consumption by weighing the remaining fuel. Record the weight and moisture content of newly collected fuel before it is added to the family's stock.
- Moisture should be read the day before from the stock of fuel that is going to be used during the next 24-hour period. For each day, randomly draw 3 pieces of the fuel and measure its moisture in three positions.

PP also ensured conservatism in seasonal variation through the timing of the survey which was aligned with the “lean period” before harvest as outlined in the KPT protocol²².

KPT Results:

A digitized data collection approach was taken by the team, where the survey responses and measurement records were recorded in the electronic survey forms through smartphones. The survey responses for each household were recorded in the centralized database, where all the survey results were combined, and the final KPT survey Excel sheet was exported.

The annual baseline fuel consumption per household was calculated from the KPT survey results of 137 households. The parameter was within the 95/10 confidence/precision which can be checked in the table below:

Mean Consumption (kg/HH/day)	12.08 (4.409 tonnes/HH/Year)
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²² <https://cleancooking.org/binary-data/DOCUMENT/file/000/000/604-1.pdf>

Standard deviation	4.67
Precision (95/10)	6.5%
Result	Ok Acceptable

Comparison with Other Projects:

In order to benchmark results derived from the project baseline KPTs and to add another layer of assurance to their validity; the below table has consolidated baseline fuel consumption values for various registered projects across Rwanda:

Registry	Project ID	Geography	Baseline fuel consumption value (tonnes/device/year)	Year of Baseline Assessment
Verra	3654	Rwanda	6.753	2020
Verra	2380	Rwanda	7.635	2021
Gold Standard	GS 11012-11027 & GS10740-10742	Western Rwanda	3.170	2021
Gold Standard	GS 10990-11006 & GS 10826-10828	Eastern Rwanda	3.760	2020
Gold Standard	GS 7449 to GS 7454	Southern Rwanda	4.470	2019
Gold Standard	GS 3444-3451, 2891-2894, 3945,3947, 1267	Eastern Rwanda	4.971	2021
Gold Standard	GS 11205	Eastern & Western Rwanda	4.037	2021 & 2022
Gold Standard	GS 11028	Western Rwanda	3.20	2021
Gold Standard	GS 11026	Western Rwanda	3.20	2021
Gold Standard	GS 10999	Eastern Rwanda	3.763	2020

The value of fuel consumption derived using baseline KPTs by the project is conservative compared to values used by other VERRA registered projects which have conducted baseline assessment during the same time under VCS²³²⁴.

The projects registered under Gold Standard have values of baseline fuel ranging from 3.17 to 4.97 tonnes/HH/year however these are region-specific data and not for the entire country. Additionally, each of the projects has used a different method for baseline assessment hence cannot be compared with the current project. A detailed excel with list of registered projects and baseline fuel consumption value has also been submitted to VVB.

Comparison with Country Data:

The National Survey on Cooking Fuel Energy and Technologies in Households, Commercial, and Public Institutions in Rwanda, a report by the Ministry of Infrastructure and Finance (Government of Rwanda), indicates that the consumption of traditional biomass is approximately 6.75 tonnes per household per year. Therefore, the applied baseline fuel consumption value of 4.409 tonnes per household per year by PP is 35% lower than the government literature's value, making it the most conservative.

This comparison, aligned with registered projects and country data, ensures that the baseline fuel consumption value is conservative, in accordance with Section 2.2.1 of the VCS Standard.

Since the grouped project activity provides improved cookstoves to households where there was only firewood was used as the cooking fuel, and the cooking device was a traditional cooking device such as three-stone fired, traditional stove without chimney etc., it is considering the burning of firewood, and usage of unimproved cooking devices as the baseline. Also, PP verifies this for every individual household where distribution of project ICS is taking place, that they were using unimproved cooking device using firewood as fuel prior to receiving it. This is acknowledged by the end-user in the form of a signed ICS usage practices poster where the information is written down. This information can be checked through the sample posters. The poster is in local language Kinyarwanda, which makes it clearly understandable for the end user.

3.5 Additionality

The Additionality for Improved Cookstoves is demonstrated using the requirements provided in the methodology VMRO006 Methodology for Installation of High Efficiency Firewood Cookstoves – version 1.1

Step 1: Regulatory Surplus

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law,

²³ <https://registry.verra.org/app/projectDetail/VCS/3654>

²⁴ <https://registry.verra.org/app/projectDetail/VCS/2380>

statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework.

The Grouped Project is a voluntary activity initiated by the project proponent. The initiative is solely dependent on VCU revenue for its sustenance and is completely financed by the project developer

Step 2: Positive List

As per the Para-7, step-2 of the methodology VMR0006, the applicability conditions described within the methodology represent the positive list. The project must demonstrate that it meets all of the applicability conditions of the methodology as well as the below condition. In so doing, the project is deemed as complying with the positive list. The project's compliance to applicability conditions has been demonstrated in section 3.2 above. In addition, the project meets the below condition:

- Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.

In case of this VCS grouped project activity, all the improved cookstoves are being distributed at zero cost to the end user.

Hence, as discussed above, distribution of Improved cook stoves within the boundaries of Rwanda may be established as additional.

Therefore, the Grouped project activity is deemed additional.

3.6 Methodology Deviations

Not applicable.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The PP plan to distribute 500,000 improved cookstove under the grouped project during first renewable crediting period. It is intended that under this GP two types of single pot portable stoves will be distributed, which are SSM S32 X stove with pot skirt and Burn Kuniokoa G2 with pot skirt. Each household will receive 1 ICS.

The first project activity instance (PAI1) has distributed 130,633 ICS, with further distributions under PAI-1 still ongoing. The first ICS under the grouped project has started distribution from 04 November 2022 and till the end of monitoring period i.e. 31 December 2022, total 130,633

stoves (wood based Kuniokoa Generation 2 with pot skirt manufactured and supplied by BURN) distributed. The below table shows distribution of stoves in November and December of 2022.

Month	Cumulative number of stoves distributed	ICS make (s)
November-2022	54,666	BURN Kuniokoa Generation 2 (with pot-skirt)
December-2022	130,633	BURN Kuniokoa Generation 2 (with pot-skirt)

The operation of stoves was confirmed by a survey undertaken as part of monitoring requirements of VCS PD and it was found that 100% of the stoves are in use as identified during the survey. The results of the survey are demonstrated in the ER spreadsheet and furnished to VVB for verification. The first monitoring survey was undertaken between 19th December and 23rd December 2022 by visiting the sampled households. Sample size was selected by using the CDM guideline, “Sampling and surveys for CDM project activities and programmes of activities” version 04. The surveyors visited the houses and completed the survey.

There was no information or event observed at the time of implementation of GP during its first monitoring period, that could impact the GHG emission reductions or removals and monitoring.

The procedures and processes set-in for the implementation of VCS GP are all in place, and do not lead to any impact on the VCU generation, or the monitoring procedures applied to calculate those.

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The applied methodology VMR0006, ver. 1.1 does not account for baseline emissions separately, rather quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves. See section 4.4 below.

5.2 Project Emissions

The applied methodology VMR0006, ver. 1.1 does not account for project emissions separately, rather quantifies emission reductions as a function of the reduction in the amount of non-renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves. See section 4.4 below.

5.3 Leakage

Leakage shall be considered as default 0.95 in accordance with section 8.3 of VMR0006.

5.4 Estimated Net GHG Emission Reductions and Removals

Equations used for Improved Cookstove as per VMR0006

$$ER_y = \sum_i \sum_j ER_{y,i,j} \quad \text{Equation (1)}$$

Where:

i = Indices for the situation where more than one type/model of improved cook stove is introduced to replace three-stone fire

j = Indices for the situation where there is more than one batch of improved cook stove of type i

ER_y = Emission reductions during year y in t CO₂e

$ER_{y,i,j}$ = Emission reductions by improved cook stove of type i and batch j during year y in t CO₂e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{woodfuel} \times (EF_{wfCO2} + EF_{wfnon-CO2}) \times N_{y,i,j} \times 0.95 \quad \text{Equation (2)}$$

Where:

$B_{y,savings,i,j}$ = Quantity of woody biomass that is saved in tonnes per improved cook stove of type i and batch j during year y

$f_{NRB,y}$ = Fraction of woody biomass that can be established as non-renewable biomass (fNRB)²⁵

$N_{woodfuel}$ = Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)

$EF_{wf\ CO_2}$ = CO₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO₂/TJ)

$EF_{wf\ non-CO_2}$ = Non-CO₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO₂/TJ)

$N_{y,i,j}$ = Number of improved cook stoves of type i and batch j operating during year y

0.95 = Discount factor to account for leakage

The quantify of woody biomass saved $B_{y,savings,i,j}$ due to implementation of improved cook stoves will be estimated as follows:

$$B_{y,savings,i,j} = B_{old} * (1 - \eta_{old} / \eta_{new,i,j}) \quad \text{Equation (3)}$$

Where:

B_{old} = Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that provided by the improved cook stove. The value of B_{old} has been sourced from baseline field surveys.

η_{old} = Efficiency of baseline cookstove

$\eta_{new,i,j}$ = Efficiency of the improved cook stove type i and batch j determined through water boiling test (WBT).

Efficiency of the improved cookstove will be updated as per equation below:

$$\eta_{new,i,y} = \eta_p \times (DFn)^{y-1} \times 0.94 \quad \text{Equation (5)}$$

Where:

η_p = Efficiency of project stove (fraction) at the start of project activity

DFn_{y-1} = Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.

0.94 = Adjustment factor to account for uncertainty related to project cookstove efficiency test.

If project households continue to use baseline cookstoves along with improved cookstoves, B_{old} shall be adjusted ex-post based on the percentage of project households found to continue such practice according to Equation 6 below. For such cases, the quantity of woody biomass saved

$B_{y,savings,i,j}$ due to implementation of improved cook stoves shall be calculated using an adjusted value to account for ex-post use of baseline stoves in addition to improved cookstove.

$$B_{old,adjusted} = B_{old} \times (1 - \mu_y) \quad \text{Equation (6)}$$

Where:

$B_{old,adjusted}$ = Adjusted Bold to account the ex-post usage of firewood in baseline cookstove(s) by project households in addition to improved cookstove (in tonnes per device)

μ_y = Baseline stove usage factor to account for use of baseline cookstoves along with improved cookstoves.

Sample VCU estimates (based on the calculation for third year of ICS monitoring estimates):

For the estimation of emission reduction, μ_y has been considered as 0.12. This is a monitored parameter and the value will be updated based on ex-post monitoring.

$$B_{old,adjusted} = 4.409 \times (1 - 0.12) \text{ tonnes/year}$$

$$\text{Or } B_{old,adjusted} = 3.880 \text{ tonnes/year}$$

The project stove efficiency as per manufacturer taken as 51.3% and 50.1.% depending on model type, which is valid if stove used with pot skirt, however, in practice it is assumed that 10% user may not use pot skirt due to various reasons/occasions, hence weighted average efficiency for year 3 is calculated as below. In this Monitoring Period only the BURN stove type has been distributed therefore efficiency of 51.3% with pot skirt is being used.

Year 3 efficiency of project stove with pot skirt- 47.26%

Year 3 efficiency of project stove without pot skirt- 39.98%

Hence, effective efficiency is calculated as

$$\text{Project stove efficiency in year 3} = 47.26\% \times 0.80 + 39.98\% \times 0.2 = 45.81\%$$

$$B_{y,savings,i,j}^{26} = 3.880 \times (1 - (0.1/0.4581)) \text{ ton firewood/household/annum}$$

$$\text{Or } B_{y,savings,i,j} = 3.033 \text{ ton firewood/household/annum}$$

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{woodfuel} \times (EF_{wf \text{ CO}_2} + EF_{wf \text{ non-CO}_2}) \times N_{y,i,j} \times 0.95$$

Further, it is assumed that 2% project might not be operational by Year-3.

$$ER_{y,i,j} = 3.033 \times 0.8872^{27} \times 0.0156 \times (112+26.23) \times (500000 \times 0.98) \times 0.95$$

$$ER_{y,i,j} = 2,708,533 \text{ ton CO}_2/\text{annum}^{28}.$$

²⁶ Assuming here that all the ICS distributed were of Kuniokoa model (bearing 51% thermal efficiency), and there was no loss in efficiency due to the non-usage of pot skirt.

²⁷NRB as per third party report for host country Rwanda submitted to VVB

²⁸ The sample calculations are for Kuniokoa ICS distribution, and ERs are calculated for its first year of operations. In the sample calculations, the accounting of discount factors such as drop-off factor and the skirt usage rate were not accounted.

Detailed calculations on the VCUs, for both (i) estimates throughout the crediting period, and (ii) actual VCUs obtained in the first monitoring period, are provided in the ER Sheet.

Emission Reduction Estimate for Improved Cookstove²⁹

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
04-November-2022 – 31-December-2022	117,023	-	-	117,023
01-January-2023 – 31-December-2023	2,136,377	-	-	2,136,377
01-January-2024 – 31-December-2024	2,708,533	-	-	2,708,533
01-January-2025 – 31-December-2025	2,595,794	-	-	2,595,794
01-January-2026 – 31-December-2026	2,555,901	-	-	2,555,901
01-January-2027 – 31-December-2027	2,542,201	-	-	2,542,201
01-January-2028 – 31-December-2028	2,508,991			2,508,991
01-January-2029 – 03-November-2029	2,027,310	-	-	2,027,310
Total	17,192,130	-	-	17,192,130

The VCS methodology, VMR0006 is applicable to both 'Projects' and 'Large Projects'. Hence there are no limits on volume of credits from Improved Cookstove component that can be certified per annum.

²⁹ For the crediting period estimates, due consideration was given to the discount factors, thus using realistic figures for the project scenario.

6 MONITORING

6.1 Data and Parameters Available at Validation

Data/parameter fixed ex-ante for Improved Cookstove instances

Data / Parameter	Bold
Data unit	tons/annum
Description	Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices
Source of data	Baseline KPTs, conducted according to KPT Version 4.0
Value applied:	4.4092 ³⁰
Justification of choice of data or description of measurement methods and procedures applied	This parameter has been fixed ex-ante. The survey revealed that the average wood consumption in the Rwanda 4.409 tonnes/device/year. The sampling was done in line with para 8.4-b of applied methodology. The details of the baseline KPTs including target population, sample size, sampling method, precision/confidence achieved has been added in section 3.4.
Purpose of Data	Calculation of emission reductions
Comments	Bold will remain fixed for the entire crediting period. The PP conducted a baseline KPT in the months of September to December 2021, to assess quantity of fuels consumed and the cooking technologies in use in pre-project scenario. The survey focused on 4 provinces (16 districts) in Rwanda. Baseline KPTs were done in 137 households to determine fuel consumption per day. The PP compared the results from the surveys with peer-reviewed publications and other registered projects and found values to be comparable/conservative.
Data / Parameter	η_p

³⁰ A baseline firewood consumption of 12.08 kg/HH/day converted into tons/hh/annum. Calculations may be checked in the ER Sheet.

Data unit	Fraction
Description	Efficiency of project stove at the start of project activity
Source of data	Certification by the Centre for Research in Energy and Energy Conservation (CREEC) lab. CREEC is a certifying agent recognized by the Rwandan Standards Bureau (RSB) ³¹ .
Value applied:	1) Kuniokoa Generation 2: • 0.4340 for project stove without skirting • 0.513 for project stove with pot skirting 2) SSM S32X: • 0.501 for project stove with skirting • 0.457 for project stove without skirting
Justification of choice of data or description of measurement methods and procedures applied	This data is fixed ex-ante as per applied methodology VMR0006.
Purpose of Data	Calculation of emission reductions
Comments	If a pot-skirt is distributed with the project stove, the parameter will be calculated as a weighted average of the efficiency of a skirted stove and an unskirted stove.

Data / Parameter	NCV _{biomass}
Data unit	TJ/ton
Description	Net calorific value of the non-renewable woody biomass
Source of data	IPCC
Value applied:	0.0156 TJ/tonne

³¹ Certification report provided to the VVB

Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	$EF_{wf\ CO_2}$
Data unit	ton CO ₂ /TJ
Description	CO ₂ emission factor for woody biomass
Source of data	IPCC
Value applied:	112 tCO ₂ /TJ
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	$EF_{wf\ non-CO_2}$
Data unit	ton CO ₂ /TJ
Description	Non-CO ₂ emission factor for woody biomass
Source of data	IPCC
Value applied:	26.23 tCO ₂ /TJ
Justification of choice of data or description of	N/A

measurement methods and procedures applied	
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	Leakage Factor
Data unit	Fraction
Description	net to gross adjustment factor to account for the leakage
Source of data	Default as per VMR0006, ver. 1.1, Section 8.3
Value applied:	0.95
Justification of choice of data or description of measurement methods and procedures applied	In-line with the applied methodology VMR0006, ver. 1.1 Section 8.3.
Purpose of Data	Calculation of emission reductions
Comments	N/A

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of woody biomass that can be established as non-renewable biomass
Source of data	Third party report in line with TOOL30 version 4.0
Value applied	0.8872

Justification of choice of data or description of measurement methods and procedures applied	The value is sourced from third party report ³² prepared in line with TOOL30 version 4 for host country Rwanda. Analysis has been added in Appendix 1 of the PD MR.
Purpose of Data	Calculation of emission reductions
Comments	The applied f_{NRB} value has been calculated by using the latest available data sources. As per the f_{NRB} calculation done by PP, the value is 91% however as a measure of conservativeness, PP used the f_{NRB} value from another registered projects which was lesser than 91%. Hence, the value has been changed to 88.72%.

6.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y
Source of data	Monitoring surveys
Description of measurement methods and procedures applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 95/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and program of activities.
Frequency of monitoring/recording	At least once every two years
Value applied	500,000 (for ex-ante estimation)
Monitoring equipment	Monitoring Surveys
QA/QC procedures applied	N/A
Purpose of data	Calculation of emission reductions.

³² Prepared by C4 EcoSolutions (Pty) Ltd dated 07-June-2023

Calculation method	N/A
Comments	This parameter only accounts for the number of project ICS that are operating in the monitoring period, and rest of the factors such as discount on non-usage of project ICS, usage of pre-project device, and drop-off are accounted separately, and hence are not reflected with the values provided for this parameter. Here, the batch j is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline stove
Source of data	Calculated using the Default value as specified in VMR0006 methodology and the data obtained through the distribution surveys
Description of measurement methods and procedures applied	If the device is a three stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney, a default value of 0.1 is used as per the applied methodology; for other types of devices, a default value of 0.2 is used.
Frequency of monitoring/recording	Fixed for each individual household at first HH visit
Value applied:	0.1
Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	Calculation of emission reductions.
Calculation method	N/A
Comments	The baseline stove is traditional stove such as three stone fired stone using firewood. Hence, efficiency of stove is used 10% which is VMR0006 default. In case there is any other unimproved

	cooking device identified as baseline cooking device, then the applied value for the parameter will be 0.2. The average of the η_{old} recorded for total project ICS households (in the distribution surveys) will be calculated for VCU calculations. For ex-ante estimates, the applied value is 0.1 (assumption for the ex-ante calculations), which will vary for each monitoring period, based on the actual data obtained at the time of distribution.
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Data / Parameter	$\eta_{new,i,j}$				
Data unit	Fraction				
Description	Efficiency of the device of each type i and batch j				
Source of data	Calculated as per the equation 5 of the applied methodology				
Description of measurement methods and procedures applied	Efficiency of the improved cookstoves will be estimated each year using equation (5) above where loss in efficiency per year is calculated, and therefore no direct measurement is involved.				
Frequency of monitoring/recording	Annually				
Value applied:	Kuniokoa ICS efficiency value for the crediting period years:				
	Crediting Period Year	Kuniokoa Efficiency value		SSM S-32 X	
		With pot skirt	Without pot skirt	With pot skirt	Without pot skirt
	Year-1	0.4822	0.4080	0.4709	0.4296
	Year-2	0.4774	0.4039	0.4662	0.4253
	Year-3	0.4726	0.3998	0.4616	0.4210
	Year-4	0.4679	0.3958	0.4570	0.4168
	Year-5	0.4632	0.3919	0.4524	0.4127
	Year-6	0.4586	0.3880	0.4479	0.4085
	Year-7	0.4540	0.3841	0.4434	0.4044
	Year-8	0.4495	0.3802	0.4389	0.4004

Monitoring equipment	N/A
QA/QC procedures applied	The values that will be applied will be based on the model of ICS distributed.
Purpose of data	Calculation of emission reductions
Calculation method	As per equation (6) above
Comments	Different models of ICS may be used at the time of distribution (hence monitoring), as indicated in Section 1.11 of PD&MR. Also, the batch j is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.

Data / Parameter	Life span
Data unit	Years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Source of data	Manufacturer specification
Description of measurement methods and procedures applied	The data source as per manufacturer specification
Frequency of monitoring/recording	Once at the time of installation of batch of type of stove
Value applied:	1) Burn Kuniokoa Generation 2 = 10 years 2) SSM S32X = 10 years
Monitoring equipment	Not applicable
QA/QC procedures applied	Not applicable
Purpose of data	Calculation of baseline emission
Calculation method	Not applicable

Comments	--
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Data / Parameter	Date of commissioning of batch j
Data unit	Date
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in “batches” and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch.
Source of data	Project database
Description of measurement methods and procedures applied	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch
Value applied:	04-November-2022 (date of commissioning of first batch of stove)
Monitoring equipment	Not applicable
QA/QC procedures applied	Not applicable
Purpose of data	Calculation of emission reduction
Calculation method	Not applicable
Comments	--

Data / Parameter	μ_y
Data unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during the year y

Source of data	Monitoring
Description of measurement methods and procedures applied	This parameter should be monitored using one of the following methods: If the baseline cookstoves are decommissioned and no longer used, as determined by the monitoring survey its value is 0 and $B_{old, adjusted}$ is equal to B_{old} . If both the improved cookstove and baseline cookstoves are used together then surveys shall be conducted to record the average continued operation of baseline cookstoves in a sample of households. The surveys should be designed to capture the cooking habits and stove usage of households in the region, including quantification of use of baseline cookstoves, by formulating questions and/or collecting evidences to determine the frequency of usage of both the improved cookstoves and baseline cookstoves.
Frequency of monitoring/recording	At least once every two years
Value applied:	0.1 ³³ (assumed for ex-ante estimation)
Monitoring equipment	N/A
QA/QC procedures applied	A 95 /10 confidence / margin of error shall be achieved for the sampling parameter irrespective of annual/ biennial monitoring frequency as per para 22 of Standard: Sampling and surveys for CDM project activities and programmes of activities, Version 09.0. In the case the desired precision is not met, lower bound will be applied.
Purpose of data	Calculation of emission reductions
Calculation method	N/A
Comments	N/A

6.3 Monitoring Plan

The monitoring plan for this project is closely derived from the methodologies. A database for the project activity will be maintained by the DelAgua.

1. Monitoring Plan Design for the Grouped Project Activity:

³³ Applied as assumption for year-1 estimations.

The monitoring plan shall consist of checking a representative sample of all ICS units distributed at least once every two years to ensure that they are still operating or are replaced by an equivalent in-service unit. A project database will be populated during the distribution of each initial cookstove issued, and will be updated based on subsequent replacements, as well as detailed data on the representative sample surveyed for monitoring purposes. The database will be accessible to the project proponent, appropriate partners, and the verification VVB. The database will include, at minimum, the following:

- Unique project identification number
- Installation date
- Family name
- Contact details of user (when available)
- Baseline cooking fuel source
- Baseline stove type
- Date of replacement of cook stove

The database will be available to select a random, representative sample for monitoring and verification purposes. This sample set will be integrated into the database to include additional monitoring parameters as required or as appropriate. Monitoring surveys will include examination of the cookstove installations and verification that they are in an operational condition. Units that have been damaged or aged beyond expected operational condition will be replaced. Therefore, should some appliances not be replaced appropriately, this will be accounted for and appropriately deducted from emission reduction claims.

1. Responsibility

The main roles and responsibilities of the parties involved is as described below:

DelAgua

- Acts as the project managing entity
- Development of PD
- Registration of the project with VERRA
- Inclusion of new PAIs upon satisfaction of the eligibility criteria stipulated in the PD
- Authorized entity for any official communication with the VERRA and VVB
- Manage and archive the database for all appliances
- QA/QC of database during distribution & data collection

- Maintain local repair/replacement facilities
- Education and public awareness raising during initial distribution as well as periodic education campaigns.
- Data collection during distribution & delivery of data to database
- QA/QC of data collection
- Conduct ongoing monitoring, as required to ascertain monitored parameters.

2. Quality Assurance and Quality Control

Spot checks of the database are conducted during the baseline surveys, distribution, and monitoring activities. Distributors are trained in the proper installation, use, and maintenance of the devices, and are provided with training sufficient to introduce recipient households to the proper operation and maintenance of the devices, as well as contact information for DelAgua. Supervisors receive similar training, as well as supervisory training in monitoring of distributor's activities. Supervisors are responsible for approximately 10 distributors. During the campaign, DelAgua staff will spot-check distributor quality, by revisiting 2 households per distributor, to verify that the stoves and /or lamps are in place, and that the recipient can show proper operation of the units. The results of this spot check are recorded electronically and compared against the initial distribution logs. The data collected in the database is verified and controlled in the following way:

1. Surveys are conducted by staff trained by DelAgua.
2. Supervisors are separately trained by DelAgua to spot-check surveyor results.
3. All ex-ante and monitored data parameters are collected on smart-phone based surveys and are uniform across all surveyors for any given survey activity.
4. Unknown to the surveyors, the database analysis includes verification of "valid" surveys including by a.) verifying household consent is received and b.) that the survey duration, as recorded by Internet time, is no shorter or longer than a reasonable value. For example, the baseline surveys for Rwanda were validated against a minimum of 20 minutes and a maximum of 120 minutes.
5. Data validation is conducted on the database to ensure that values are numerically consistent, i.e. multiple choice values add up to 100% of respondents.

3. Training and maintenance procedures:

DelAgua is responsible for training any contractors used during distribution, education and monitoring activities. DelAgua ensures training of all on-site staff with respect to adherence to the Monitoring Plan of the project activity. Records of the training will be kept. Internal audit of

all the records of the distribution, monitoring and education activities are performed on an annual basis, and will be carried out once (or twice, based on the requirements and plans) a year. Training records are saved for such training programs, where step-by-step training is provided to the surveyors for a quality data collection process, where they are expected to not impact the materially, degree of conservativeness, and perform the surveys in an unbiased manner. The surveyors are trained on a periodic basis, and refresher trainings are also provided to improve the quality of surveys from time to time. During these audits all the data and parameters that need to be monitored as per the monitoring plan will be checked and shortcomings if any will be reported and addresses. Since the surveys are electronic in format, recording of responses and the record keeping procedure for these surveys are conveyed to the team through rigorous trainings.

4. Calculation of emission reductions:

Emission reductions will be calculated using the results of the most recent survey data. The updated parameters adjust all emission reduction results for the year monitored.

5. Sampling Plan:

STEP 1: Monitoring Objectives

The objective was to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of monitoring period.

- a) $N_{y,i,j}$ Number of project devices of type i and batch j operating during year y (Annual/biennial)
- b) μ_y Adjustment to account for any continued use of pre-project devices during the year y

As per Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities version 9.0, 95/10 confidence/precision has been adopted as this is a large-scale project for both the parameters.

STEP 2: Target Population

Total 130,633 households (end-users) who were enrolled in the project between 04-November-2022 to 31-December-2022 were considered for sampling.

STEP 3: Sampling Frame

The sampling frame referred to all the information sources on the database. PP collected all the information sources on monitoring data through physical visit. The monitoring survey data was collected primarily through physical survey which is the primary mechanism in place for the data collection of the newly distributed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) which is used in the current monitoring period.

Following information were recorded during the monitoring survey.

- a. Product based Unique Identification number.
- b. Contact information, geographical location or other relevant information of the end user
- c. Sampling frame was representative of the entire population
- d. No elements from outside the population of interest is present for the current monitoring period. Wherever information miss-match is observed, it is corrected in the database and conservative assumptions is applied.

STEP 4: Sample Method

The sampling method for the monitored parameters $N_{y,i,j}$, and u_y is Simple Random Sampling and samples were randomly selected from the primary sampling units as illustrated above.

To ensure a random selection of ICS, random number generators was applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame.

Applying the random number generators, the ICS was then be randomly chosen from the defined population up to the required sample size as calculated by the project proponent.

To determine the parameters, sampling involves the following approaches

$N_{y,i,j}$: Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)

$\mu_{y,i,j}$: Pre project device only is in use then fraction to be used to calculate total number, however if pre project device is used along with project ICS, proportion of usage of each was determined by cooking habits evaluated by survey questionnaire during the monitoring period. Using the formulas below, the project proponent randomly sampled the required number of ICS from the primary sampling units

STEP 5: Desired Precision / Expected Variance and Sample Size

To estimate the number of premises to be surveyed with a 10% margin of error at desired confidence level of 95%, the optimal sample size n is given by:

1. For proportional parameter i.e. number of operational project cookstoves

$$n \geq \frac{1.96^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.96^2 p(1-p)}$$

Here,

Parameter	Description
-----------	-------------

n	Sample Size
N	Total number of cookstoves (ICS) distributed
p	Expected proportion
1.96	Represents the 95% confidence required
0.1	Represent the 10% relative precision

2. For mean-based parameter i.e adjustment factor

$$n \geq \frac{1.96^2 NV}{(N-1) \times 0.1^2 + 1.96^2 V}$$

Here,

Parameter	Description
n	Sample Size
N	Total number of cookstoves (ICS) distributed
SD	Standard deviation
V	(SD/mean) ²
1.96	Represents the 95% confidence required
0.1	Represent the 10% relative precision

Requirements of Guideline for Sampling and surveys for CDM project activities and program of activities, version 4.0 has been followed.

STEP 6: Minimum Sample Size

- **Proportion based parameter ($N_{y,1,1}$)**

As per the CDM guideline for Sampling and survey³⁴, if the sample size calculation returns a value of less than 30 samples, a minimum sample size of 30 shall be chosen. For the current monitoring period the sample size came to be 9. The sample size calculation is shown below:

Sampling Constants	Values
Monitoring period start	04-11-2022
Monitoring period end	31-12-2022
Level of Sampling	Single vintage
Confidence (%) (90 or 95)	95%

³⁴ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20151023152925068/Meth_GC48_%28ver04.0%29.pdf

Margin of Error (%)	10%
Z value	1.960

Usage Rate ($N_{y,i,j}$)				
Year	Stove Model (Sampling Frame)	Stove Population (sampling frame size)	Usage Rate ($N_{y,i,j}$) (%)	Calculated Sample Size (n)
2022	Kuniokoa	130,633	80%	96
Sample size determination				
Estimated $N_{y,i,j}$ (p)				80.00%
$V_{ny,i,j} = p(1-p)$				0.160
Sample Size required ($N_{y,i,j}$)				96
Monitoring Results				
Year	Stove Model (Sampling Frame)	Sampling frame size	Monitored Sample Size	Monitored Usage
2022	Kuniokoa	130,633	120	100%
Reliability Check				
Samples Monitored				120
Monitored $N_{y,i,j}$ (p)				100.00%
Standard Error of $N_{y,i,j}$				0.00%
Relative precision (Margin of error)				0.00%
Result				Ok, acceptable
Lower Bound confidence value				not applicable

As per the sample size calculation, 96 samples were calculated to be surveyed, However, 120 samples were considered to compensate for any outliers or non-responses associated with samples. The parameter was within the 95/10 confidence/precision.

- **Mean based parameter (μ_y)**

If the parameter of interest is a numeric mean value (i.e. not a proportion or percentage) the Student's t-distribution shall be used if the resulting sample size is less than 30. As per the CDM: Sampling and Surveys for project activities and programme of activities, PP used t-distribution to calculate the sample size.

Use of baseline stove (μ_y)					
Year	Stove Model (Sampling Frame)	Stove Population (sampling frame size)	Expected Mean value (tonnes per year)	Expected SD	Calculated Sample Size (n)
2022	Kuniokoa	130,633	0.1	0.01	7
Sample size determination					
Estimated (p)					0.10
Estimated Standard Deviation (SD)					0.01
$V_{mean} = (SD/p)^2$					0.01
Minimum Sample Size required					4
tDistribution sample size adjustment				Iteration 1	11
				Iteration 2	5
				Iteration 3	8

				Iteration 4	6
				Iteration 5	7
				Iteration 6	6
Monitoring Results					
Year	Stove Model (Sampling Frame)	Sampling frame size	Monitored Sample Size	Monitored Mean Value of use of baseline stove Monitored Standard Deviation	
2022	Kuniokoa	130,633	120	0.0629	
Reliability Check					
Samples Monitored(p)					120
Mean value of μ_y					0.0629
Standard error of μ_y					2.22%
Relative precision (Margin of error) (%)					0.58%
Result					ok acceptable
Applicable Value					not applicable

As per the sample size calculation, 7 samples were calculated to be surveyed, However, 120 samples were considered as per CDM sampling standard. There were two reasons- firstly, common survey was conducted for $N_{y,i,j}$ and μ_y and secondly, to compensate for any outliers or non-responses associated with samples. The parameter was within the 95/10 confidence/precision.

6. Quality Assurance/Quality Control

The project proponent applied a confidence/precision of 95/10 for each monitored parameter allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The choice of measure applied to both parameters i.e. number of operational cookstoves, and adjustment factors. The project proponent monitored a particular parameter till the required level of confidence/precision has been reached, as long as the calculated minimum number of samples has been achieved.

The monitoring plan followed procedures in place to ensure good quality data. The project proponent ensured that field personnel have reviewed, understood and have agreed to follow the monitoring plan procedures, including provisions for maximizing response rates, documenting out-of-population cases, refusals and other sources of non-response. A quality control and assurance strategy were audited and documented. The monitoring plan explained how to properly survey households to prevent bias from interviewer. In the case a household refuses to participate, another household is chosen at random. To reduce interviewer bias, good questionnaire design and well-tested questionnaires was developed. The calculation of the sample size was carried out using values from pilot study for parameter proportions (for number of operational cookstoves), mean values (for adjustment factor) and standard deviations, as the actual characteristics of the population/sampling frame are unknown.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type i and batch j operating during year y
Value applied:	130,633
Comments	Measured directly or based on a representative sample. Sampling standard has been used for determining the sample size to achieve 95/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and program of activities. For the current monitoring period the usage rate of the ICS is 100%.

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline stove
Value applied:	0.1
Comments	A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood, or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney

Data / Parameter	$\eta_{new,i,j}$
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Data unit	Fraction		
Description	Efficiency of the device of each type i and batch j		
Value applied:	Crediting Period Year	Kuniokoa Efficiency value	
		With pot skirt	Without pot skirt
	Year-1	0.4822	0.4080
Comments	The value applied is 0.4769 as weighted average considering pot skirt usage rate 92.78% as per sample survey results. Since M.P.-1 accounts for year-1 only, the applied values indicate year-1 values for Kuniokoa ICS model.		

Data / Parameter	Date of commissioning of batch j
Data unit	Date
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in “batches” and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch.
Value applied:	04-November-2022 (distribution of first ICS)
Comments	-

Data / Parameter	Life span
Data unit	Years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Value applied:	Burn Kuniokoa – 10 years

Comments	As per manufacturer specification. During the current monitoring period, only Burn Kuniokoa has been distributed.
Data / Parameter	μ_y
Data unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during the year y
Value applied:	0.0629
Comments	A 95 /10 confidence / margin of error has been achieved for the sampling parameter irrespective of annual/ biennial monitoring frequency as per para 22 of Standard: Sampling and surveys for CDM project activities and programmes of activities, Version 09.0. The desired precision/confidence was met for the parameter during the current monitoring period. Total sample monitored for this parameter was 120 as PP conducted common survey for parameter $N_{y,i,j}$ and μ_y .

7.2 Baseline Emissions

Not applicable. The applied methodology VMR0006 does not have provision to calculate baseline emissions as separate.

7.3 Project Emissions

Not applicable. The applied methodology VMR0006 does not have provision to calculate project emissions as separate.

7.4 Leakage

A net to gross adjustment factor of 0.95 is already part of the methodological equation to calculate the Emission Reductions. The methodology states that there is no requirement to monitor leakage as separate if this net to gross adjustment factor is applied for the ER calculations.

7.5 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project. Since ICS is replacing the baseline traditional stoves using non-renewable biomass, therefore equations 1 and 2 of the methodology have been applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j}$$

Equation (1)

Where:

i	Indices for the situation where more than one type/model of improved cook stove is introduced to replace three-stone fire
j	Indices for the situation where there is more than one batch of improved cook stove of type i
ER_y	Emission reductions during year y in t CO ₂ e
$ER_{y,i,j}$	Emission reductions by improved cookstove of type i and batch j during year y in tCO ₂ e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{wood\ fuel} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

---Equation (2)

Where:

EF_{wf,CO_2}	CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO ₂ /TJ) ³⁵
$EF_{wf,nonCO_2}$	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO ₂ /TJ)
$N_{y,i,j}$	Number of improved cook stoves of type i and batch j operating during year y
0.95	Discount factor to account for leakage

³⁵ https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf

$B_{y,savings,i,j}$	Quantity of woody biomass that is saved in tonnes per improved cook stove of type i and batch j during year y
$f_{NRB,y}$	Fraction of woody biomass that can be established as non-renewable biomass (fNRB) (Calculated using Tool 30)
$NCV_{wood fuel}$	Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)

The quantity of woody biomass saved $B_{y,savings,i,j}$ due to implementation of improved cook stoves has been estimated and fixed ex-ante by the following equation:

$$B_{y,savings,i,j} = B_{old} * (1 - \eta_{old} / \eta_{new,i,j}) \quad \text{.....Equation (3)}$$

B_{old}	Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that provided by the improved cook stove. The value of B_{old} can be sourced from historical data or baseline surveys.
η_{old}	Efficiency of baseline cookstove
$\eta_{new,i,j}$	Efficiency of the improved cook stove type i and batch j determined through water boiling test (WBT)

$$\eta_{new,i,y} = \eta_p \times (DF_n)^{y-1} \times 0.94 \quad \text{.....Equation (5)}$$

Where:

η_p	Efficiency of project stove (fraction) at the start of project activity
$(DF_n)^{y-1}$	Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.
0.94	Adjustment factor to account for uncertainty related to project cookstove efficiency test.)

Where the project households continue to use baseline cookstoves along with improved cookstove cookstoves, B_{old} is to be adjusted ex-post based on the percentage of project

households found to continue such practice according to Equation 6 of VMR0006 methodology. For such cases, the quantity of woody biomass saved $B_{\text{savings},i,j}$ due to implementation of improved cook stoves shall be calculated using an adjusted value to account for ex-post use of baseline stoves in addition to improved cookstove.

$$B_{\text{old adjusted}} = B_{\text{old}} * (1 - \mu_y) \quad \text{Equation (6)}$$

Where:

Bold, adjusted	Adjusted <i>Bold</i> to account the ex-post usage of firewood in baseline cookstove(s) by project households in addition to improved cookstove (in tonnes per device)
μ_y	Baseline stove usage factor to account for use of baseline cookstoves along with improved cookstoves

For example, considering one ICS/household, the saving for the entire monitoring is calculated as:

$$B_{y,\text{savings},i,j} = B_{\text{old adjusted}} * (1 - \eta_{\text{old}} / \eta_{\text{new},i,j})$$

$$= B_{\text{old}} * (1 - \mu_y) * (1 - \eta_{\text{old}} / \eta_{\text{new},i,j})$$

$$B_{y,\text{savings},i,j} = 4.409 * (1 - 0.0629) * [1 - (0.10/0.4769)]$$

$$= 3.2654 \text{ tonnes}$$

$$ER_{y,i,j} = B_{y,\text{savings},i,j} * f_{\text{NRB},y} * NCV_{\text{wood fuel}} * (EF_{\text{wf},\text{CO}_2} + EF_{\text{wf},\text{non CO}_2}) * N_{y,i,j} * 0.95$$

$$= 3.2654 * 0.8872 * 0.0156 * (112 + 26.23) * (130,633 * 100\%) * 0.95$$

$$= 59,473 \text{ tCO}_2\text{e (Rounded down)}$$

Year	$\eta_{\text{new},i,j}$ (efficiency of cookstove)	$B_{y,\text{savings},i,j}$ (Quantity of wood saved)	$ER_{i,j}$ (Emission reduction in tCO ₂)
1	47.69%	3.2654 Tonnes	59,473

The methodology does not account for project emissions separately, but instead quantifies net GHG emission reductions achieved by this project

Leakage is considered as default 0.95 in accordance with Section 8.3 of the VMR0006 methodology.

Therefore, the annual emission reductions considering total 130,633 ICS distributed in PI1 are estimated in the table below.

The VCU's obtained during M.P.-1 are provided in the table below. The detailed calculations for this may be referred from the ER Sheet.

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
04-November-2022 – 31-December-2022	59,473	-	-	59,473
Total	59,473	-	-	59,473

Comparison between ex-ante estimates and the actual ERs obtained during the monitoring period:

<u>Ex-ante emissions reductions /removals</u>	<u>Achieved emissions reductions /removals</u>	<u>Percent difference</u>	<u>Justification for the difference</u>
117,023 ³⁶	59,473	-49.18%	The emission reduction achieved during current monitoring period are ~49% lower than value estimated for same period is mainly due to difference in the operational days. During ex-ante estimation, emission was calculated considering 58 days whereas ex-post the average operational days is 28 considering stoves were distributed during the monitoring period.

³⁶ See ER Sheet, worksheet “ER Estimates Summary”, cell D4 for calculations.

APPENDIX 1: FNRB CALCULATION

Step 1: Estimation of renewable biomass (RB) in the country of Rwanda as per equation (4) of the fNRB tool

$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other,i})) \quad \text{Equation (4)}$$

Where:

- $MAI_{forest,i}$ = Mean Annual Increment of woody biomass growth per hectare in sub-category i of forest areas in the relevant period (tonnes/ha/yr)
- $MAI_{other,i}$ = Mean Annual Increment of woody biomass growth per hectare in sub-category i of other land areas in the relevant period (tonnes/ha/yr)
- $F_{forest,i}$ = Extent of forest in sub-category i in the relevant period (ha)
- $F_{other,i}$ = Extent of other land in sub-category i in the relevant period (ha)
- $P_{forest,i}$ = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (in sub-category i) in the relevant period (ha)
- $P_{other,i}$ = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other land areas (in sub-category i) in the relevant period (ha)
- i = Sub-category i of forest areas and other land areas³

Parameters used:

Input parameter	Value	Data source
Extent of forest and other wooded land in 2021 (in ha), $F_{forest,i}$	467,239	Geospatial data products for Rwanda were analysed to estimate Rwanda's renewable biomass. The forest and other wooded land cover for 2000 and 2021 was estimated using Hansen/UMD/Google/USGS/NASA ³⁷ spatial data for 2000 and 2021, which is derived from Hansen et al. ³⁸ , and disaggregated according to the FAO ³⁹ global ecological zones. The total woody cover extent, including forests and other wooded areas, was calculated for each ecological zone, within the protected areas and within areas that are either accessible or geographically remote.
Protected areas within forest areas (in ha)	137,289	
Remote areas within forest areas (in ha)	12,681	

³⁷ Hansen/UMD/Google/USGS/NASA. Hansen Global Forest Change v1.9 (2000-2021). Available at: <https://glad.earthengine.app/view/global-forest-change>. Accessed on: 31 May 2023.

³⁸ Hansen, M. C. C. et al. High-resolution global maps of 21st-century forest cover change. Science (80- .). 342, 850–854 (2013)

³⁹ FAO. 2012. Global ecological zones for FAO forest reporting: 2010 Update. Forest Resources Working Paper 179. Food and Agriculture Organisation of the United Nations, Rome. Available at: <https://www.fao.org/3/ap861e/ap861e00.pdf>.

Extend of non-accessible area (in ha), $P_{forest,i}$	149,970	Calculated (protected forest + remote forest)
Mean Annual Increment (t/ha/year), $MAI_{forest,i}$	1.82	The mean annual increment for forest is calculated as default age weighted average of the forest type and as per IPCC ⁴⁰ values for Tropical Mountain system.

Geographically remote areas were determined to be areas beyond a harvestable distance from roads and settlements. Conservatively, it was assumed that forests and wooded areas adjacent to all roads within a distance of 2.5 km are harvestable, despite how geographically remote or inaccessible these roads may be. This threshold was determined based on the peer-reviewed literature on wood harvesting practices.

Banks et al.¹⁸ found that woody biomass density increases significantly as a function of distance from the edge of a settled area. Beyond 450 m, average biomass density was found to increase from 2.1 to 16.6 t/ha. Jumbe and Angelsen⁴¹ modelled fuelwood choice from household survey data. They found that the most significant determinants of fuelwood choice are source attributes (size and species composition) and distance to source. The mean harvesting distance for their study was found to be 0.95 km. Bandyopadhyay et al.⁴² matched household survey and remote sensing data to investigate wood fuel use in relation to poverty levels. They concluded that the one-way walk duration to wood fuel source for both poor and non-poor households is 0.6 hours. Assuming an average walking speed of 4 km/hr over uneven terrain, the average one-way distance to wood fuel source is less than 2.5 km. Areas beyond the harvestable distance were, therefore, determined to be geographically remote. Accessible areas are those that are not protected and not geographically remote, from which the annual renewable biomass production is calculated.

The default age-weighted mean annual increment (MAI) estimates of each ecological zone, as reported by the IPCC, were used for this study. The proportion of forest stand ages above and below 20 years old, including primary forests, were estimated for each ecological zone by extrapolating the observed forest gain extents between 2000 and 2012 to a 21-year period. The average MAI estimates for Rwanda is 1.82 t/ha/year for tropical mountain system between 2000 and 2012 to a 20 years period.

FAO Global Ecological Zone	Total forest cover (ha)	Protected area cover (ha)	Remote area cover (ha)	MAI (t/ha/yr)	Annual growth, RB(t/yr)
Tropical mountain system	467,239	137,289	12,681	1.82	578,495
Total	467,239	137,289	12,681	-	578,495

Total renewable biomass RB = 578,495 t/year

⁴⁰ IPCC. Forest Land. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4: Agriculture, Forestry and Other Land Use 4, 1–29 (2019)

⁴¹ Jumbe, C. B. L. & Angelsen, A. Modeling choice of fuelwood source among rural households in Malawi: A multinomial probit analysis. Energy Econ. 33, 732–738 (2011).

⁴² Bandyopadhyay, S., Shyamsundar, P. & Baccini, A. Forests, biomass use and poverty in Malawi. Ecol. Econ. 70, 2461–2471 (2011)

Step 2: Estimation of consumption of woody biomass (H) in the country of Rwanda as per equation (2) of the fNRB tool

$$\text{NRB} = \text{H} - \text{RB}$$

Where,

H = Total consumption of woody biomass in the applicable area in the relevant period (tonnes)

$$\text{H} = \text{HW} \times \text{N} + \text{CE} + \text{NE}$$

Where,

HW = Average consumption of wood fuel per household, including fuelwood and charcoal, in the applicable area in the relevant period (tonnes/household)

N= Number of households consuming wood fuel within the applicable area in the relevant period (number)

CE= Commercial woody biomass consumption for energy applications (e.g. commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

NE= Commercial woody biomass consumption for non-energy applications (e.g. construction, furniture) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

The total woody biomass consumption combines the country's domestic and non-domestic *energy* and non-domestic *non-energy* consumption. Rwanda's estimate of total woody biomass consumption estimation is derived from World Bank population statistics⁴³ and consumption data sourced from literature^{44,45}, the FAO⁴⁶ and KPT surveys. Reported values of charcoal consumption are converted to the equivalent woody biomass using guidelines on kiln efficiency from the FAO forestry paper on industrial charcoal making⁴⁷ and the latest CDM Tool 30 v04.0 2022 default charcoal-to-wood biomass conversion factor. The CDM default value of 4 was applied to the urban domestic charcoal consumption and non-domestic energy charcoal consumption.

Data was drawn from the most recently available official reports/statistics, in this case from the UN's Statistics Division to demonstrate both the recent trend in increase of wood fuel use at the household level, plus also the most recently available data (from 2021). This increase in fuelwood use at the household level has had a significant impact on the parameter value fNRB. The density of fuel is not a time bound parameter, hence average of Africa is used as per published data by FAO.

Input parameter	Value	Data source
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43 <https://data.worldbank.org/>.

44 United Nations & African Union. Review of woodfuel biomass production and utilization in Africa

45 FAO. Forestry Sector in 2020: <https://www.fao.org/forest-resources-assessment/fra-2020/country-reports/en/>.

46 FAO Statistics. Forest Products 2019. <https://www.fao.org/3/cb3795m/cb3795m.pdf> (2021)

47 Charcoal transition: <https://www.fao.org/3/i6935e/i6935e.pdf>

Wood fuel consumption (m ³ /capita/yr)	0.396	FAO ⁴⁸
Population of the country	13,276,517	World Bank ⁴⁹
Wood fuel consumption (m ³ /yr)	5,257,220	Calculated
Conversion factor (t/m ³)	0.725	FAO ⁵⁰
HW (Average consumption of wood fuel tonnes/yr)	3,811,384	Calculated

Input parameter	Value	Data source
H (in t/year)	6,177,233	Calculated
HW woody biomass consumption in households (in t/year)	3,811,484	Calculated
CE Non domestic woody biomass consumption for energy application (in t/year)	201,877	Calculated, using wood to charcoal conversion factor ⁵¹
NE Non domestic woody biomass consumption for non-energy application (in t/year)	2,163,872	Calculated using biomass expansion factor 1.50
Population of the country	13,276,517	World Bank ⁵²

Equation (3) of the fNRB tool stipulates that the average household wood fuel consumption is multiplied with the number of households and industries consuming wood fuel for energy and non-energy applications within the country. Since most of the available and reliable data is total wood fuel consumption in households and commercial and other use.

The total population of host country has been sourced as per the published World Bank population data from 2022. This is in line with the fNRB tool.

Step 3: Estimation of quantity of non-renewable biomass (t/year) as per equation (2) of the fNRB tool

$$\text{NRB} = \text{H} - \text{RB}$$

Where,

⁴⁸ <https://www.fao.org/faostat/en/#data/FO>

⁴⁹ <https://data.worldbank.org/>

⁵⁰ <http://www.fao.org/3/a-i6935e.pdf>

⁵¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-33-v2.0.pdf>

⁵² <https://data.worldbank.org/>

H = Total consumption of woody biomass in the applicable area in the relevant period (tonnes)

This step calculates the quantity of non-renewable biomass (t/year) (NRB) by subtracting the renewable biomass estimated in step 1 from the total consumption of woody biomass estimated in step 2. No further input parameters are needed under this step.

$$NRB = 6,177,233 - 578,495$$

$$NRB = 5,598,738 \text{ t/year}$$

Step 4: Estimation of fraction of woody biomass that can be established as non-renewable (fNRB) as per equation (1) of the fNRB tool

$$fNRB = \frac{NRB}{NRB + RB} \quad \text{Equation (1)}$$

This step calculates the fraction of woody biomass that can be established as non-renewable (fNRB). The non-renewable biomass calculated in step 3 is divided by the sum of non-renewable biomass and renewable biomass calculated in step 1. No further input parameters are needed under this step.

$$fNRB = 578,495 / (5,598,738 + 578,495)$$

$$fNRB = 91.0\%$$

Step 5: Comparison as per para 6 of CDM Tool 30 version 4

As per the Bailis report, the f_{NRB} value for Rwanda is 59%. Rwanda is considered as wood fuel “hotspot” as most of the harvested wood fuel is unsustainable. The resulting f_{NRB} value is higher than the expected f_{NRB} values determined in Bailis Report⁵³ using the WISDOM method. While direct comparisons between the WISDOM and CDM methodologies are only sometimes appropriate—given the different approaches of the methodologies—the following factors can contribute to variations in the f_{NRB} estimates between the application of the CDM method in the present study and the WISDOM method in the Bailis Report⁵⁴:

- **Parameter H (Total consumption of woody biomass in the applicable area in the relevant period (tonnes)):** The current report uses the most recent population statistics (2021) for Rwanda that represents an increase in population numbers since the 2009 data utilized in the Bailis Report. Hence, the value of parameter H is higher than the one used Bailis report as it is calculated using the total population.
- Updated 2019 FAO forest products statistics were used in the present study, whereas the comparison study used 2013 FAO forest products statistics.
- **Parameter RB (renewable biomass):** The approach used to determine the amount of forest that is accessible yields lower estimates in the present study than the comparison study. This results in a lower estimated RB and consequently increased NRB and f_{NRB} .

⁵³ Bailis R, Drigo R, Ghilardi A and Masera O. 2015. The carbon footprint of traditional woodfuels. Nat Clim Chang 5: 266–272

⁵⁴ Ibid.

- **Parameter MAI (Mean Annual Increment):** In the present study, MAIs were calculated using a weighted average based on the forest area of three categories (i.e., primary forests, above 20-year secondary forest, below 20-year secondary forest). Data from the 2019 Refinement of the IPCC Guidelines was used in combination with extrapolating the observed forest gain extents between 2000 and 2012 to a future 21-year period. As per the Bailis Report, MAI values were derived from a combination of field observations and IPCC values, followed by a different estimation of growth rates as a percentage of standing stock. This approach often yields higher MAIs and may lead to higher estimations of RB and subsequently, lower estimations of NRB and fNRB.

The calculated fNRB value of 0.91 indicates that the consumption of woody biomass from the accessible forest in Rwanda is greater than the country's capacity to renew the supply sustainably. This finding is supported by the consistent and widely-reported deforestation rate and dynamics observed for the country^{55,56,57}.

The fNRB value of the project was also compared against other registered projects across VERRA and Gold Standard in Rwanda. The fNRB value used by recently registered projects are as follows:

Registry	Project ID	fNRB
Verra	2380	97%
Gold Standard	4261	98%
Gold Standard	2450	98%
Gold Standard	11205	90%
Gold Standard	2449	98%
Gold Standard	11098	96.82%
Gold Standard	12308	88.72%

The fNRB value used in other registered projects across various standards range from 88.72% to 98%. As a measure of conservative, the fNRB value for the project has been changed to 88.72% which is the lowest among all the registered projects.

Step 6: Cross check in line with para 13 of CDM Tool 30 version 4.0

In line with para 13 of TOOL 30 V4, the NRB was compared to the top-down product of average above-ground biomass (146.06 t/ha) and the most recent annual deforestation data (3,187 ha/yr). NRB, as calculated in this report according to the CDM Tool 30 v4.0 2022, was found to be 5,598,738 t/yr, which is greater than the cross-check results based on deforestation (465,493 t/yr). It is expected that the cross-check estimate of biomass loss from deforestation

55 FAO. 2020b. Global Forest Resources Assessment 2020: Main Report. Food and Agriculture Organisation of the United Nations, Rome. Available at: <https://www.fao.org/forest-resources-assessment/2020/en/>

56 Global Forest Watch (GFW). 2023.

57 Mongabay. 2020.

could underestimate the extent of non-renewable biomass as it only considers the average biomass stocks from completely deforested areas. Unsustainable wood harvesting resulting in forest degradation and substantial reductions in biomass stocks, but not complete deforestation is not considered in the cross-check. Consequently, the estimate of NRB is considered applicable for this study given that the guidance provided by CDM Tool 30 v4.0 2022 was followed.